



# A review of literature on dropout in tertiary-level continuing professional development e-learning courses

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## ABSTRACT

The increasing importance of lifelong learning and professional development is linked to the growing usage of tertiary-level continuing professional development (CPD) through e-learning. However, high dropout rates remain a significant challenge both for adult learners seeking to update their skills and for institutions aiming to provide effective online education.

This mapping review analyses 216 articles on dropout in e-learning courses for tertiary-level continuing professional development, published between 2007 and 2023. The review addresses three key research questions: what has been researched regarding dropout, how the research has been conducted, and what findings have emerged concerning dropout rates, causes, and interventions.

The review finds that research in this field is dominated by quantitative methods and the focus is largely on MOOCs and on courses in information and communication technology (ICT), business, and STEM subjects. High dropout rates are common, though they varied widely, based on differences in how dropout was defined and measured. Time constraints, insufficient motivation, and course workload were cited more frequently as contributing factors. Persistence is associated with motivational support, high-quality course design, effective feedback, and social interaction. Interventions such as encouragement and guidance showed some success, but remain understudied.

The review highlights a need for more qualitative methodologies, theoretical frameworks, and research on interventions, particularly outside the MOOC context and in professional/commercial adult education. To improve retention, practitioners should focus on flexible course structures, social features, assessment and feedback implementations, and approaches that foster motivation and self-regulation.

## 1. Introduction

The need for highly educated workers has increased around the world, making continuing professional development (CPD) more important than ever [1]. There are several groups of actors who provide CPD courses. Universities and other higher education institutes organise their own continuing education and open education courses [2, p. 14]. Commercial training companies provide such training most often in the form of short courses with a typical duration between one and three days. Many non-profit organisations provide online courses and learning resources for self-studying.

Courses can be organised as classroom courses, online courses (i.e. distance education, e-learning), or in blended form combining both on-site and online activities. A significant share of online education in the last decade has been organised as massive open online courses, MOOCs, which may enrol tens of thousands or more participants, though many courses attract much smaller numbers. In contrast, a

Small Private Online Course (SPOC) is an online course designed for a limited group of participants, often offering more interaction and personalised support than large-scale MOOCs [3]. SPOCs are often designed by (or in partnership with) technology companies to prepare learners for industry certification exams, such as those offered by Microsoft, Cisco, AWS, and SAP [4].

A major challenge in online education is *high dropout*, which has been a major problem for decades and as such a subject of multiple studies [5–8]. We have carried out an extensive review of 216 articles that address the dropout phenomenon, and our findings confirm this challenge, but also show a more complex picture. For traditional online courses, dropout is a significant problem, with medians around 37%–55%, although the limited number of studies with reported values cautions against drawing strong conclusions about specific patterns. The issue is more acute in MOOCs, where median dropout rates exceed 80%. However, these figures require careful interpretation: as discussed

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in Section 5.6, calculation methods affect reported rates, although our findings indicate that MOOC dropout remains consistently high even when excluding individuals who never engaged with content. These high dropout rates have sparked significant interest in researching the factors behind these numbers, predicting student success in online courses, and developing pedagogical interventions to reduce dropout.

Previous reviews of e-learning dropout have mainly focused on MOOCs in general, specific learner groups such as university students, programme level dropout, or technical approaches such as dropout prediction, but none have systematically examined dropout specifically in the context of *continuing professional development*. This distinction matters because the dynamics of dropout in CPD differ from those in general higher education. Unlike undergraduate students, whose retention is often driven by the external pressure of degree completion and academic grades, CPD learners are typically voluntary participants motivated by immediate career advancement or specific skill acquisition [9,10] [1, p. 62]. They face unique time constraints due to balancing full-time work and family obligations, and they often engage with content selectively, consuming only what is immediately relevant [9] [11, p. 129]. This pattern of selective learning challenges the traditional definition of dropout: for a CPD learner, leaving a course after achieving a personal learning goal constitutes success, yet is often categorised as failure in research that does not specifically account for the adult learning context [12,13]. There is also a gap with respect to the courses offered by commercial providers, which constitute a substantial part of the CPD ecosystem but remain almost entirely absent from the research literature. Our primary contribution is to conduct the first systematic mapping study of dropout specifically within the tertiary-level CPD e-learning domain. A key strength of this methodological approach is its ability to reveal the activity foci of the field: by systematically classifying and counting contributions, we can precisely identify areas of intense research focus (the “overstudied”), as well as significant gaps (the “understudied”).

To address these gaps, the objective of this review is to synthesise and structure the research landscape on dropout in tertiary-level CPD e-learning. Specifically, our goal is to uncover: (1) the topics and questions that have been investigated, (2) the research methods and contexts that have been used, and (3) the key findings regarding dropout definitions, contributing factors, and interventions.

To achieve this, the review is guided by the following research questions.

- RQ 1 What has been researched regarding dropout in tertiary-level continuing professional development e-learning courses?
- RQ 2 How has the research been carried out? What have been the research contexts, target groups, and research methods?
- RQ 3 What results related to dropout have been found?
  - RQ 3.1 How has dropout been defined and how should dropout percentages be interpreted from this perspective?
  - RQ 3.2 What have been found as the reasons for dropout and what kind of interventions have been carried out to diminish it?

To answer these questions, we conducted a systematic mapping study, with the specific procedures for this methodology detailed in Section 4.

The importance of research in this area lies in the high dropout rates observed in online tertiary-level CPD e-learning courses. High dropout not only limits individual opportunities for skill development, but also impacts the larger goals of lifelong learning and workforce competency, especially with respect to changing workplace requirements [11]. By clarifying the factors underlying dropout and identifying effective pedagogical approaches and interventions, this review provides guidance to support learner persistence and success in studies, whether that means fully completing a course or achieving individual learning goals.

The main audience for this review is providers of continuing and adult education, including universities, higher education institutions, and commercial training organisations, as well as teachers and course designers involved in online learning for adults. The findings also offer implications for policymakers and others working on adult lifelong learning. Researchers studying dropout may find the suggested further research and other results useful for their own work.

The paper is structured as follows. We first discuss related review studies identified during our literature search. We continue to present the terminology and scope in Section 3 and the review method in Section 4. The results are presented in Section 5. Finally, we discuss the implications of the findings in Section 6.

## 2. Previous review studies

Dropout in online learning is clearly a significant issue, as discussed in several previous review studies. To position our contribution and justify the need for this review, we examined the reviews identified during our systematic search (see Section 4). We found that these existing reviews do not specifically address the focus of our study: dropout in tertiary-level Continuing Professional Development (CPD) e-learning.

In a well-cited article by Lee and Choi [8] on online post-secondary course dropout, the background was on how online course dropout is problematic for several audiences. Quoting:

For students, a failure in or failure to complete the first online course may lead to lower self-esteem and self-confidence, and discourage from registering for further online courses. For institutions, high dropout rates suggest that online programmes are ineffective and of poor quality. [8, p. 594]

A large portion of the reviews we found, detailed in Table 1, concentrated on MOOCs. Some offered very broad overviews of MOOC research, where dropout was just one topic among many. Others focused specifically on MOOC dropout itself, examining aspects such as high dropout rates, engagement patterns, research methods, or contributing factors. Although these reviews cover courses often undertaken by adult professionals for CPD purposes, their primary lens was typically the general MOOC phenomenon rather than a systematic analysis focused specifically on the unique characteristics, motivations, and constraints prevalent in the tertiary-level CPD e-learning context.

Another group of reviews focused on technical methods, especially dropout prediction using data science and machine learning [14–17]. These articles are useful for understanding predictive modelling techniques, but naturally offer less insight into the wider pedagogical landscape, dropout definitions, or interventions outside the prediction context.

Finally, some reviews targeted specific factors, such as engagement, motivation, or gamification [18–20]. Although providing depth on those topics, they lacked the comprehensive scope needed to systematically map the general landscape of dropout causes, contexts, methods, definitions, and interventions.

Next, we discuss the foci and findings of the reviews listed in Table 1.

In a mapping study of MOOCs by Rasheed et al. [21], the inspiration was that there had not been a generalised review study that covers the entire MOOC landscape to provide a big picture of MOOCs, since MOOCs became mainstream about a decade ago before the article. In that paper, dropout was identified only as one area of MOOC research without any closer investigation of it. Ejreaw and Drus [22] conducted a review with the dual objective of learning more about MOOC platforms and their use to improve the educational process and study the challenges of participants that lead them to drop out of the courses, with an added focus on institutional challenges. Bezerra and Silva [23] directly aimed to identify the reasons for the high

**Table 1**  
Review articles.

| Article name and year                                                                                                 | Target years           | Article count      | Focus                                                                                                                                     |
|-----------------------------------------------------------------------------------------------------------------------|------------------------|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| A review of literature on the reasons that cause the high dropout rates in the MOOCs (2017) [23]                      | 2008–2016              | 24                 | Dropout definition, reasons for MOOC dropout, categorisation as MOOC or student related, discussion of possible solutions                 |
| A survey of machine learning approaches for student dropout prediction in online courses (2020) [15]                  | –2019                  | ≤79 <sup>a</sup>   | Dropout prediction with machine learning: student modelling, prediction strategy taxonomy, evaluation measures, datasets, open challenges |
| A systematic mapping study of the empirical MOOC literature (2019) [21]                                               | 2009–2018 <sup>b</sup> | 311                | Systematic mapping of empirical MOOC research; methods, focus, publication trends and venues, contributions; future research              |
| Completion factor in massive open online course in developing countries: A literature review in 2015–2021 (2022) [24] | 2015–2021              | 89/26 <sup>c</sup> | MOOC dropout study methodology, target countries in research, retention factors                                                           |
| Educational data science in massive open online courses (2017) [14]                                                   | 2012–2015              | ≤61 <sup>a</sup>   | MOOC challenges and educational data science: student interaction analysis, dropout prediction, feedback, adaptivity                      |
| Engagement and desertion in MOOCs: systematic review (2022) [18]                                                      | –2021                  | 40                 | Engagement and dropout in MOOCs; analyses key variables and challenges; future research                                                   |
| Gamification increases completion rates in massive open online courses (2022) [20]                                    | –2022 <sup>d</sup>     | Over 70            | MOOC dropout and persistence reasons, gamification in MOOCs                                                                               |
| MOOC dropout prediction using machine learning techniques: review and research challenges (2018) [16]                 | –2018 <sup>d</sup>     | ≤40 <sup>a</sup>   | MOOC dropout prediction with machine learning: dropout factors; prediction methods, challenges, and solutions                             |
| Prediction in MOOCs: a review and future research directions (2018) [17]                                              | 2014–2018 <sup>e</sup> | 88                 | MOOCs used for prediction, predicted outcomes, prediction features, techniques and models, result evaluation; future research             |
| Take a MOOC and then drop: a systematic review of MOOC engagement pattern and dropout factor (2023) [25]              | 2014–2021              | 21                 | MOOC engagement patterns and dropout factors in higher education, pedagogical implications; future research                               |
| The challenges of massive open online courses (MOOC) – a preliminary review (2017) [22]                               | –2017 <sup>d</sup>     | ≤21 <sup>a</sup>   | MOOC platform features, dropout rates and factors, institutional challenges, potential solutions                                          |
| The role of motivation in MOOCs' retention rates: a systematic literature review (2022) [19]                          | 2015–2020              | 50                 | Motivational factors and theories for MOOC completion; future research                                                                    |

<sup>a</sup> Count of references, no direct figure of review result count given.

<sup>b</sup> First ones found in 2011.

<sup>c</sup> 89 = review results from all over the world, 26 = from developing countries.

<sup>d</sup> Data collection year not reported; review's publication year used.

<sup>e</sup> Start year not reported; year of earliest included article used.

dropout rates in MOOCs. There were three purposes in the review by Liliana et al. [24]: what methods are typically used in dropout studies, which countries conduct most of these studies, and what factors affect retention in MOOCs? The part related to dropout in the review by Estrada-Molina and Fuentes-Cancell [18] analysed the variables most studied in participation to reduce the dropout rate in MOOCs, and similarly in the review by Huang et al. [25] there was an objective to know what factors affect MOOC dropouts in higher education. Badali et al. [19] focused on motivation in relation to MOOC dropouts, both from a theoretical and practical perspective. The impact of gamification on MOOC completion rates was analysed by Nesterowicz et al. [20], comparing dropout in courses with and without gamification.

Some review articles were related to the mathematical analysis of the dropout phenomenon, either from a data science point of view, as in the review by Romero and Ventura [14], or with the subject area of dropout prediction. The prediction of dropout was divided into machine learning techniques, as in the reviews by Prenkaj et al. [15] and Dalipi et al. [16], or through more general predictive approaches as discussed by Moreno-Marcos et al. [17].

This review differs from the existing literature in the following key aspects.

**Previous reviews** Although previous reviews have addressed important questions about dropout in MOOCs, often focusing on specific aspects such as dropout factors or prediction techniques (as summarised in Table 1), this study adopts a broader perspective as the research target. Our aim is not to synthesise answers to specific questions but rather to comprehensively chart the research landscape within the tertiary-level continuing professional development (CPD) e-learning context.

**Primary studies** A gap is apparent in the primary research literature, as in the 216 studies retrieved, only ten focus on dropout related to adult education as a specific research objective. This confirms that the unique challenges of CPD learners are under-researched.

**Contributions from the systematic mapping methodology** This review addresses the identified gap by providing what is, to our knowledge, the first application of systematic mapping methodology specifically to the domain of dropout in tertiary-level CPD e-learning, covering both university continuing education and commercial adult education courses. With 216 articles, our synthesis is also more extensive than most prior reviews on this topic, which have generally been narrower in scope or based on a smaller number of studies (see Table 1). Unlike broader, non-dropout-specific MOOC mappings [21] or reviews on specific factors, our methodology allows for a comprehensive charting of the research landscape within this particular domain. We analyse research topics, contexts, methods, dropout definitions, dropout/persistence factors, and interventions relevant to CPD, including university-based continuing education courses and seeking evidence from the commercial sector (confirming its near-absence in the literature). By systematically classifying the articles, we clearly identify both researched areas and important under-researched gaps, such as non-MOOC CPD, commercial courses, qualitative approaches, and theory development, giving a solid foundation for future work.

3. Terminology and scope

To clarify the scope of this article, we define the following **key terms**.

**Formal education and training** is defined by the OECD as planned education provided in the system of schools, colleges, universities, and other formal educational institutions [26, p. 1].

**Non-formal education** is defined as a sustained educational activity that does not correspond exactly to the above definition of formal education. Non-formal education may therefore take place both within and outside educational institutions. It may cover, for example, education programmes in life skills, work skills, and general culture. Educational activities can include open or distance learning courses, private lessons, organised sessions for on-the-job training, and workshops or seminars. Some of these learning activities might have a short duration [26, p. 1].<sup>1</sup>

**University-based education** refers to formal degree-orientated programmes and courses offered directly by universities or higher education institutions, which typically lead to academic credit or qualifications [27].

**Adult education and training** has been defined by the OECD as referring to adults’ participation in formal and/or non-formal education and training [1, p. 113], and we use that definition in this article. We recognise that there have also been other views and definitions; for example, in the United Kingdom it has had a connotation to mean liberal education [28, p. 44], and UNESCO has also included informal education to it [29, pp. 6–7].

**Tertiary education** builds on secondary education, providing learning activities in specialised fields of education. It aims at learning at a high level of complexity and specialisation. Tertiary education includes what is commonly understood as academic education, but also includes advanced vocational or professional education [27, p. 69].

**Continuing professional development** has been defined in multiple ways over the years [30]. In this study, we use it to refer to the systematic enhancement of professionals’ competencies throughout their working lives, more specifically on non-formal learning activities undertaken voluntarily. We deliberately avoid the term “continuing (professional) education”, which in some contexts is associated with training dictated by law or licencing requirements (see [31] as an example).

**Commercial e-learning** is used in this review specifically to refer to for-profit organisations that directly provide courses to learners or employers, often focused on industry certifications (for example, the certifications related to Cisco, SAP, or Microsoft). This represents a subset of the broader commercial e-learning ecosystem described by Williamson and Hogan [32].

**Massive open online courses (MOOCs)** are a specific format of online courses characterised by open enrolment, a large number of participants, and flexible access. MOOCs can be created by universities, commercial providers, or collaborations between institutions and may serve both university-based and CPD purposes depending on their design and target audience [14,23].

To further clarify the specific educational landscape addressed in this review, a comparison of three core intersecting concepts of adult education (defining the learner), continuing professional development (defining the purpose) and commercial e-learning (defining a key provider context) is presented in Table 2.

Table 2  
Comparison of core concepts.

| Aspect             | Adult education                                                                                        | Continuing professional development (CPD)                                                              | Commercial e-learning                               |
|--------------------|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|-----------------------------------------------------|
| Defined by         | Learner demographic                                                                                    | Purpose of learning                                                                                    | Provider and business model                         |
| Brief definition   | Adults pursuing learning after initial schooling                                                       | Career-focused skill enhancement for professionals                                                     | Online courses provided by for-profit organisations |
| Scope              | Broadest category, encompassing any learning by adults (formal, non-formal; personal, or professional) | Subset of adult education focused on professional competencies                                         | Delivery model that is a subset of CPD              |
| Typical motivation | Can be career-driven, but also for hobbies, or obtaining a formal degree                               | Upskilling, reskilling, maintaining professional certification, or career advancement                  | Gaining specific skills or industry certifications  |
| Examples           | Adult taking evening classes in leather-working, adult returning to university for a master's degree   | Java developer taking a course on Spring Boot, project manager getting a Microsoft Azure certification | SAP training course from a private provider         |

4. Review method

4.1. Mapping studies

The methodology for this type of overview has been described variously in the literature as “systematic mapping studies”, “scoping studies”, “mapping reviews”, and “scoping review” [33,34]. A recent article by Campbell et al. [34] helps to clarify these terms by distinguishing between “mapping reviews”, broad overviews using pre-defined frameworks to classify a large body of literature, and “scoping reviews”, which may be more exploratory and in-depth on a narrower topic.

Following the established methodology in our field, we refer to our approach as a *systematic mapping study*. The idea of a mapping study has been described as follows [33, p. 1]:

Systematic mapping studies or scoping studies are designed to provide an overview of a research area through classification and counting contributions in relation to the categories of that classification.

Compared to systematic reviews, there are differences related to research questions, search process and strategy, quality evaluation, and handling of results [33, p. 2], [35, p. 44], [36, p. 7].

First, the research questions of mapping studies are typically broader and there are multiple of them. Second, the search terms are not as focused as in systematic reviews, and consequently more results are returned. This is not a negative aspect as the aim is for broad coverage. Moreover, it is not imperative to find all existing articles as the interest is in research trends. Third, in systematic mapping studies, no quality assessment is needed. For example, articles with no empirical evidence can be important in detecting trends in research topics.

Extracting data in mapping studies typically consists of classification and categorisation, with sufficient details to answer the research questions but without being a too time consuming endeavour. Similarly, the analysis phase is concerned with summarising the data to answer the posed research questions, rather than including in-depth analysis techniques.

The search process follows the guidance given in [37, p. 19], [35, p. 49], [38, p. 5], and it is depicted in Fig. 1. We explain each step in more detail below.

<sup>1</sup> The OECD categorisation also includes *informal learning*, which refers to unstructured learning from daily activities. It is not within the scope of this article.



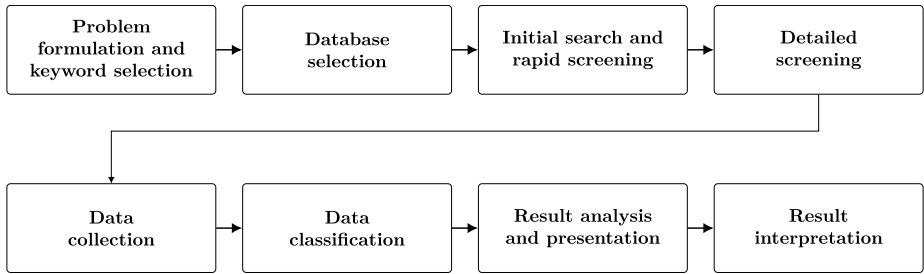


Fig. 1. Search steps.

Table 3  
Search keywords and notes of the literature review.

| Database                    | Keywords                          | Notes                              |
|-----------------------------|-----------------------------------|------------------------------------|
| ACM                         | +e-learning +dropout              | Journals and conferences           |
| ScienceDirect               | “E-learning” AND Dropout          |                                    |
| IEEE Xplore                 | Education AND Dropout             |                                    |
| ERIC (ProQuest)             | (Electronic Learning) AND Dropout | Scholarly (peer reviewed) journals |
| SAGE journals               | “e-learning” AND dropout          |                                    |
| EBSCOhost APA               | e-learning AND dropout            |                                    |
| PsycArticles                |                                   |                                    |
| Scopus                      | “e-learning” AND dropout          | No patents or citations            |
| Google Scholar <sup>a</sup> | “MOOC” “dropout”                  |                                    |

<sup>a</sup> Unlike the others, Google Scholar is a search engine, not a managed database.

4.2. Problem formulation and keyword selection

As the research questions concerned various different aspects of dropout, a very precise set of search keywords could not be extracted, which fits the typical characteristics of a mapping study [35, p. 44]. As a result, we decided to use rather generic keywords and use more time in the search and screening phases to gather as good a coverage of the subject as possible.

The keywords used for the search were “e-learning” and “dropout”, and were adjusted according to thesaurus of each database, if available. The exact keywords used are in Table 3. The limitations from this keyword choice are discussed in Section 4.6. As a special note, “MOOC” was used instead of “e-learning” in Google Scholar due to an enormous amount of 55 700 hits with the latter.

4.3. Database selection

We decided to conduct the search in several databases of journals that are shown in Table 3. The databases are shown in the order of the search; thus, for example, ACM Digital Library was searched completely before moving on to ScienceDirect. The articles were checked for duplicates already at this phase. Thus, if an article was already found in an earlier database in the database list, it was not included in the search results of a later database. As a consequence, the numbers given by this step do not include the duplicates.

4.4. Initial search and rapid screening

The study selection process followed a systematic approach, intended to generate a comprehensive collection of relevant literature identifiable through the specified search parameters rather than a random sampling of the field. Thus, the resulting analysis focuses on describing the characteristics of this set of articles rather than estimating population parameters with confidence intervals.

Some methodological sources for literature reviews suggest a separate phase for an initial search, followed by a rapid screening phase (for

a collection of processes, see [35, p. 49]). The initial search immediately showed a very large number of result articles, and simultaneously a high percentage of articles that were not relevant. As examples of non-relevant articles, there were many that discussed “dropout” as a regularisation technique in machine learning algorithms, a concept unrelated to student withdrawal from education. Several studies addressed dropout at the primary or secondary school levels (ISCED [27] levels 1 to 3), which were also outside the scope of this review. For these reasons, the initial search and rapid screening phases were combined, as also suggested in [38, p. 52]. Similarly, the articles returned by the initial search, which were not relevant, were not noted in a results spreadsheet.

In total, 5888 articles were returned in this phase.<sup>2 3</sup> The database searches were executed sequentially between September 2023 and May 2024. Consequently, the literature coverage for each individual source reflects the status of that database at the time of its specific search. No publication date restrictions were imposed during the process; the resulting time span of 2007–2023 reflects the actual publication years of the relevant literature identified during these searches. The full selection process is illustrated in Fig. 2 as a PRISMA diagram [39].

To ensure that the final set of articles is aligned with this review’s scope, we used certain inclusion and exclusion criteria. The rationale for these criteria is as follows.

- Course-level vs. programme-level dropout: We only included studies targeting course-level dropout and excluded those focusing on programme-level dropout. This distinction is relevant because the factors influencing a learner’s decision to leave a single course (e.g., content difficulty, acute time constraints, lack of interest in some parts of the content) are often different from the reasons for abandoning an entire degree or multi-year programme (e.g., long-term financial issues, career changes, or institutional barriers). Our focus is therefore on the internal dynamics of individual e-learning courses.
- Educational context: The review is centred on tertiary-level CPD. Therefore, studies at the K-12 level were excluded as the pedagogical context and learner motivations are fundamentally different. Furthermore, to isolate factors specific to adult learners, we excluded courses populated exclusively by traditional university students undertaking formal degree programmes, unless the course was a MOOC (which typically attract a wide, adult participant demographic). This ensures that our findings are relevant to the challenges faced by adult professionals in balancing education with work and life commitments.

<sup>2</sup> The Google Scholar query did return a very large amount of papers, 6270. For this reason, we decided to stop going through the results when finding new relevant studies was getting very scarce; this happened at 510 studies. At this point, the last 20 results did not show new inclusions.

<sup>3</sup> The Scopus search returned a large number of articles in which the word “dropout” was only in the references. A programmatic solution was used to filter these out.

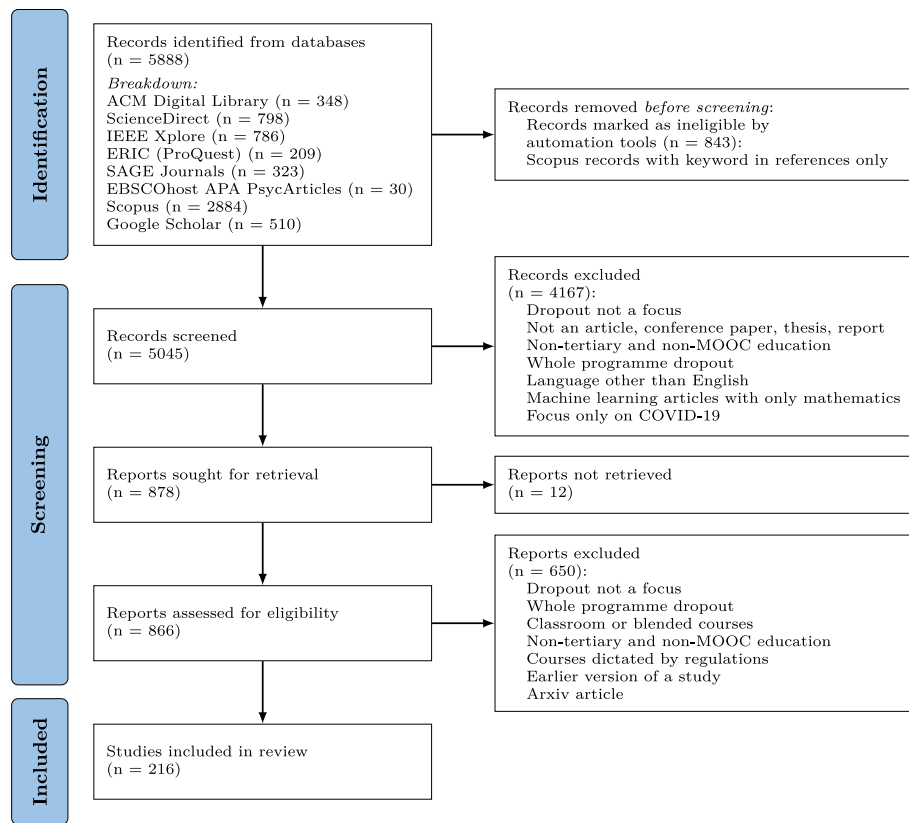


Fig. 2. PRISMA flow diagram of the selection process.

- Course delivery modes: We included only fully online e-learning courses to isolate variables related to the online environment. Classroom and blended courses were excluded because face-to-face components add variables such as in-person social dynamics and direct instructor presence that would complicate the analysis of factors unique to distance learning.
- Relevance of machine learning studies: Many articles use “dropout” as just a binary number that they could try to predict. We included machine learning papers only if they provided information on the phenomenon of dropout (for example, by identifying key predictive factors). Studies that focused solely on the mathematical treatment or algorithmic performance of prediction models were excluded, as our goal is not to review prediction techniques but to inspect findings on the causes of and interventions for dropout.
- Ensuring finality and quality: To base our review on the most complete and peer-reviewed version of a given study, we excluded earlier versions (for example, a conference paper that was later expanded into a journal article) and preprints from sources like ArXiv. This prevents redundancy and ensures that this review is based on the most mature version of the research.
- COVID-19-specific studies: We excluded articles in which the primary focus was on the impact of the COVID-19 pandemic on dropout rates. Although important, such studies analyse an anomalous period and the factors identified (e.g., sudden transition to remote mode of studying, lockdown-related stress) may not be generalisable to the typical e-learning context.
- Legally required courses: We excluded studies where the courses were legally compulsory for participants, such as training required for professional licencing. Motivation to complete a mandatory course is driven by external regulatory pressure, rather than intrinsic interest or voluntary professional development goals. Including such studies would obscure the factors that

influence dropout among voluntary adult learners, which are the focus of this review.

Based on the rationale above, the selection rules applied during the rapid screening phase are depicted in Table 4. These rules focused on criteria that could be quickly assessed from the title and abstract, such as educational level or machine learning context. Other criteria from our rationale, such as the exclusion of blended or classroom courses, were often not evident until a more thorough reading and therefore were primarily applied during the detailed screening phase (see Table 6).

We first looked at the title of the articles. In many cases, it was immediately clear that the paper did not deal with learner dropout, but rather with something else. As discussed above, for example, the term “dropout” could have been used in the context of machine learning, or then some rapid screening rule such as the K12 level of education, or being part of a programme, could be seen from the title alone.

If the titles did not reveal enough information for the filtering decision, the abstracts of the studies were consulted to verify whether the study would be relevant. If the abstract did not clearly show relevance, the full text was consulted around the keyword “dropout”.

A further rule we followed was that borderline cases were accepted.

Rapid screening yielded 866 articles from the separate databases as shown in Table 5. The screening was carried out by the first author alone, because for most criteria the include/exclude decision was straightforward.

#### 4.5. Detailed screening

The detailed screening phase had inclusion and exclusion criteria as presented in Table 6.

In this study, no methodological quality screening was used to limit the number of articles, which is typical of mapping studies [36, p. 7].

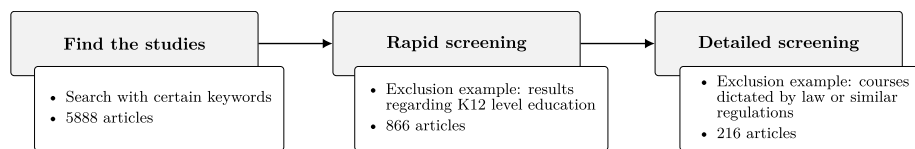


Fig. 3. Search, rapid and detailed screening.

Table 4

Rapid screening selection rules.

| Rule category                      | Inclusion rule                                                                                                                                                                                                                                  | Exclusion example                                                                                                 |
|------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Article coverage regarding dropout | The article had to actually be about learning dropout. Example: There could be a long discussion about dropout, but the discussion would mostly use terms that were used to operationalise dropout research, like “engagement” or “motivation”. | The word “dropout” might be found only in the reference list of the paper, and not handled in the actual contents |
| Document type                      | Research articles, conference papers, theses and reports                                                                                                                                                                                        | Posters, correspondence and commercial books                                                                      |
| Educational level                  | Studies where the level of education is tertiary education, and all MOOC related studies                                                                                                                                                        | K12 education related studies                                                                                     |
| Course/programme dropout           | Studies targeting course level dropout                                                                                                                                                                                                          | Studies covering whole programme dropout                                                                          |
| Language                           | Studies in English                                                                                                                                                                                                                              | Studies written in other languages                                                                                |
| Article accessibility <sup>a</sup> | Studies that were accessible                                                                                                                                                                                                                    | Studies behind a paywall from our institute                                                                       |
| Machine learning                   | Machine learning articles that tell more about the dropout phenomenon, for example, by describing the most important predictors of dropout                                                                                                      | Machine learning articles, which only contain mathematical treatment (most often prediction) of dropout           |
| COVID-19                           | Articles that only mentioned COVID-19                                                                                                                                                                                                           | Articles focusing specifically on COVID-19                                                                        |

<sup>a</sup> A total of 12 articles were sought but not retrieved due to paywall restrictions. While four of these appeared potentially relevant based on their titles or abstracts, the remainder were identified as likely of limited relevance. Given the high volume of retrieved literature, the exclusion of these few reports is not expected to influence the study’s overall findings.

This phase was performed mainly by the first author. The omission or inclusion of 17 borderline cases was discussed with the second author, which resulted in omitting six studies. This approach was chosen over a larger set of papers checked by both authors, as the criteria were quite clear to enforce, and consequently there was a rather minimal number of papers requiring more thorough discussion.

The final count after this phase was 216 articles. Article counts of all phases can be found in Fig. 3. As the total number of relevant articles is quite high, continuing the search with snowballing by studying the references of articles [38, p. 29] or finding citing articles [37, p. 87] were not used to further increase the number of articles.

#### 4.6. Limitations

Only the word “dropout”/“dropouts” was used in the keyword search. Although related terms such as “attrition”, “persistence”, or “retention” are common in the literature, our search strategy focused on the full text of documents rather than titles and abstracts alone. Consequently, relevant articles primarily using these synonyms were

Table 5

Database hit counts and unique rapid screening results.

| Database                   | Hit count   | After rapid screening |
|----------------------------|-------------|-----------------------|
| ACM Digital Library        | 348         | 81                    |
| ScienceDirect              | 798         | 69                    |
| IEEE Xplore                | 786         | 94                    |
| ERIC (ProQuest)            | 209         | 51                    |
| SAGE journals              | 323         | 16                    |
| EBSCOhost APA PsycArticles | 30          | 13                    |
| Scopus                     | 2884        | 396                   |
| Google Scholar             | 510         | 146                   |
| <b>Total</b>               | <b>5888</b> | <b>866</b>            |

Table 6

Detailed screening selection rules.

| Rule category                      | Inclusion rule                                                                                                                     | Exclusion example                                                                                                                   |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Article coverage regarding dropout | The article had to actually be about learning dropout                                                                              | Dropout as a minor mention only, with no real results or discussion                                                                 |
| Course/programme dropout           | Studies targeting course level dropout <sup>a</sup>                                                                                | Studies covering whole programme dropout, or intention to continue using MOOCs in the future                                        |
| Course type                        | E-learning courses                                                                                                                 | Courses having face-to-face components, essentially classroom or blended <sup>b</sup> courses                                       |
| Participant demographics           | MOOCs, and courses having a large portion of adults as participants where the course is not part of a formal educational programme | Courses having only university students as participants                                                                             |
| Regulations                        | The target courses are not compulsory in a legal sense, as such a setting would affect dropout                                     | Courses dictated by laws or similar regulations, for example some medical courses                                                   |
| Article version                    | The paper is the only version, or the final version in a chain of different versions                                               | The paper has a direct continuation article, for example, a conference paper was omitted if it had also had a later journal version |
| Article source                     | Article is not an Arxiv article <sup>c</sup>                                                                                       |                                                                                                                                     |

<sup>a</sup> This criterion was already included in the initial search/rapid screening phase. It is repeated here as a more thorough perusal of study contents revealed some articles to actually be about whole programme dropout.

<sup>b</sup> A course having combination of distance and classroom education elements.

<sup>c</sup> This criterion could have been utilised already at the initial search/rapid screening phase. However, it was established after internal discussion after that phase had been completed.

likely still captured if the term “dropout” appeared even once in the manuscript. This keyword restriction was used for the following reasons.

- An exceptionally large number of studies was already found using these keywords, and adding synonyms would have primarily increased noise (false positives) rather than relevant hits

- An earlier study [40] using all keywords yielded only a few more documents not already included in the results obtained with the “dropout(s)” keyword
- The search was carried out on eight different databases to ensure a broad coverage of the literature
- Given the large scale of studies identified, increasing the number of search terms would have resulted in a volume of studies beyond the scope and aim of this review

As is typical in mapping studies, this review did not include an assessment of the quality of the articles. Therefore, the findings reflect the factors and patterns that were found, but the results have not been weighted or prioritised based on the methodological quality of individual studies.

#### 4.7. Data collection

To systematically answer the research questions, the final set of articles was classified according to several criteria. The information gathered can be described through the following broad categories. They will be described more thoroughly in the rest of the section.

- **To answer RQ 1 (What has been researched regarding dropout in tertiary-level continuing professional development e-learning courses?),** we collected data to map the research landscape, identify its main areas of focus, and classify the types of questions being asked. This included:
  - *Bibliographic data:* To analyse publication trends and identify influential authors and articles (as presented in Section 5.1), we collected publication year, author information, and citation counts for each article
  - *Analysis of literature focused on adult learners:* To provide a detailed qualitative summary of the most relevant literature, we first identified the special subset of studies ( $n = 10$ ) that explicitly targeted adult learners as their main focus of research. As this learner demographic is central to the CPD domain, we performed an analysis of their main findings, which are presented in Section 5.2.
  - *Classification of research questions:* To create a comprehensive quantitative overview of the entire field, we extracted the research questions from all 216 articles and developed a data-driven classification scheme for those. The classification categories include, for example, prediction, causes, or interventions. These are presented in Section 5.3.
- **To answer RQ 2 (How has the research been carried out? What have been the research contexts, target groups, and research methods?),** we gathered information on the methodological approaches and contexts of the selected studies. This involved classifying:
  - *Research design and methods:* To understand how the research was conducted, we categorised the research design (quantitative, qualitative, etc.), the use of any theories, models, or similar, used as frameworks in the studies, and the specific data analysis methods employed. These findings are in Section 5.4.
  - *Data sources and context:* To describe the setting of the research (presented in Section 5.5), we collected these types of information:
    - \* *Data source setup and data types:* These contain the primary data source (for example, direct course data, ready-made datasets, questionnaires) and the kinds of data used (for example, clickstream, recorded activities, or demographics)

- \* *Course characteristics:* Details on the course subject, educational level, reported workload, duration, and number of course participants
- \* *Participant demographics and course profiles for MOOCs:* We gathered specific data from MOOC studies to characterise their relation to adult education. This included reported participant demographics (age, employment, education) as well as course design details: prerequisite information and the intended target group. This focused analysis was not needed for the other online courses, which were already selected for their explicit adult learner focus. This analysis allowed us to assess whether MOOCs can serve as a reasonable proxy for other adult CPD courses, given the small amount of research in those contexts.

- **To answer RQ 3 (What results related to dropout have been found?),** we extracted the key findings from the literature. Our data collection was divided into two main components to address the sub-questions:

- **For RQ 3.1 (How has dropout been defined and how should dropout percentages be interpreted from this perspective?),** we collected data on how the concept of dropout was defined, measured, and interpreted (Section 5.6). This included:

- \* *Operational definitions:* To understand the mechanics of dropout measurement, we identified the temporal starting point (for example, from enrolment vs. first activity) and the criteria used to define a student as a dropout or as having passed
- \* *Reported dropout rates:* We collected the actual dropout percentages to analyse how these rates varied based on the operational definition used and by course type
- \* *Learner typologies:* We identified articles that discussed learner intent and offered alternative conceptualisations, such as different types of learners (for example, casual learners, samplers, or completers)

- **For RQ 3.2 (What have been found as the reasons for dropout and what kind of interventions have been carried out to diminish it?),** we extracted findings related to the causes of dropout and the strategies used to mitigate it (results in Sections 5.7, 5.8, and 5.9). This contained the following:

- \* *Dropout and persistence factors:* We extracted and categorised all factors that the articles associated with either dropping out (e.g., time constraints) or persisting (e.g., provided support). This also included identifying and discussing the categorisation frameworks used in the literature to group these factors. Furthermore, we analysed the factors identified within the subset of studies categorised as employing primarily qualitative methodologies, to highlight insights potentially different from the quantitative trends.
- \* *Interventions and their effects:* We identified studies that implemented an intervention, collecting data on both the type of intervention (for example, gamification) and its reported effect on dropout
- \* To supplement the main analysis from all retrieved articles, we performed a thematic analysis focused on the most impactful publications. We identified core journals ranked in the first quartile (Q1) within their fields that contained at least three articles from our final set. This step allowed us to compare the findings from the leading publications with the trends in the entire dataset.



In addition to collecting data directly related to the research questions, we systematically extracted and categorised two further types of information synthesis presented within the reviewed articles:

- Pedagogical implications: We collected recommendations for teaching practice, course design, or policy suggested by the original authors based on their findings (presented in Section 5.10)
- Suggestions for further research: We gathered the directions for future work proposed by the authors of the reviewed articles (Section 5.11)

Also, special notes of the research were gathered, where appropriate. As an example, it was noted whether an article contained a separate discussion of the definition of dropout.

There were numerous cases in which the first author was not himself sure of how to classify a certain feature. These cases were then classified by the second author and then discussed together until consensus was reached. The unclear cases were marked on the classification spreadsheet as comments.

Most of the data collected were qualitative in nature. Only the number of subjects (persons) in a study was typically reported as an exact number. The course workload also had numerical values, for example, a duration of certain weeks. Unfortunately, any duration values do not really give much information about the actual workload, because taking a course may request only a few minutes or several hours of work every day. Only in a few cases was a somewhat comparable figure identified, for example, an ECTS value<sup>4</sup> [41], a number of hours [42,43], or minutes [44,45]. Thus, the workload cannot be treated as a quantitative value.

The main author was listed for each article; more specifically, the first author mentioned was considered the main author. This allowed us to find out whether some authors would be notably influential in the field given a high number of articles.

The research questions were extracted and qualitatively classified. This classification analysis was data-driven and, therefore, the step was performed all over again after all papers were processed once. Then, analysing the initial categories showed that many of them would actually represent a similar feature, and could thus be combined.

The research settings were classified according to the largest set of variables in this study, e.g. data set size, study type (quantitative/qualitative/mixed), what kind of data sources were utilised, and were interventions used. The actual content of each course was classified by its topic according to the *ISCED categorisation of the fields of education and training* [46]. The educational level was categorised based on the course context, using the labels “university-based MOOCs” and “non-formal education MOOCs” for MOOCs, and “university” and “non-formal education” for regular online courses.

In addressing the definition of dropout, the data collected from the articles included (1) how dropout was operationalised for measuring, such as if dropout percentages were based on the total number of enrolments or only on the students who actually started the course, (2) the criteria used for course completion or passing, (3) the reported dropout percentages and their categorisation (for example, by ranges like 0%–20%, 20%–40%, etc.), and (4) how dropout was understood as a concept, for example, by introducing a category of casual learners who pick and choose only what is important to them. The analysis also included finding differences between MOOCs and other online courses and analysed how these affect dropout rates.

Regarding dropout and persistence factors, the findings considered features that were identified or discussed as related to dropout or persistence, such as time constraints, motivation and its subcategories, course workload and quality of the e-learning environment.

<sup>4</sup> European Credit Transfer System. 1 ECTS credit corresponds to approximately 27 h of total work, including self-study.

**Table 7**  
Mapping of research questions to result sections.

| Research question | Result subsection(s)                                                                             |
|-------------------|--------------------------------------------------------------------------------------------------|
| RQ 1              | 5.1 Bibliographic analysis<br>5.2 What has been researched<br>5.3 Research questions in articles |
| RQ 2              | 5.4 Research design, theories, and analysis methods<br>5.5 Data sources                          |
| RQ 3.1            | 5.6 Dropout definition                                                                           |
| RQ 3.2            | 5.7 Dropout and persistence factors<br>5.8 Interventions                                         |

In analysing interventions, the findings included the type and description of the intervention (such as communication strategies, guidance, gamification, peer feedback or rewards), the observed effect of the intervention on dropout or persistence, and whether the effect of the intervention was positive, neutral, or negative.

Finally, going beyond typical systematic mapping studies, we did analyse which themes regarding dropout were included in the most impactful publications in our set of articles. We included sources ranked in the first quartile (Q1) within their fields.<sup>5</sup> We also added a threshold for including publications that had at least three articles in our set. This resulted in the selection of eight core journals for the thematic analysis: *Computers & Education*, *Education and Information Technologies*, *Distance Education*, *Computers in Human Behavior*, *IEEE Access*, *International Journal of Educational Technology in Higher Education*, *IEEE Transactions on Learning Technologies* and *Journal of Computing in Higher Education*. In total, 37 articles were included in these publications.

4.8. Data analysis

The analysis was primarily descriptive and quantitative, aiming to map the research landscape rather than to conduct a meta-analysis, thus we employed descriptive statistics to synthesise the extracted data. For categorical variables (for example, course topics, research methods, intervention types), we calculated frequencies and relative percentages to identify dominant trends and gaps.

For the primary outcome variable, *dropout rates*, we calculated averages (mean and median) and spread (range and standard deviation) to summarise the distribution of values reported in the primary studies. These calculations treat each study as a single unit of analysis. We attempted to calculate participant-weighted averages to account for the variance in course sizes; however, this was not feasible because only a small fraction of studies reported both the dropout rate and the participant count.

Given the diversity of the included studies regarding target groups, course lengths, and definitions of dropout, calculating statistical correlations or combining results into a meta-analysis was not feasible.

5. Results

The results to the research questions are reported as shown in Table 7. In addition to these main findings, we collected and categorised both the pedagogical implications and the further research directions suggested, as proposed by the original studies. These sections reflect the instructional recommendations and open research questions highlighted by the field itself (see Sections 5.10 and 5.11, respectively). A more thorough analysis of research gaps and recommendations based on our own synthesis of the literature is provided separately in the Discussion (Section 6).

<sup>5</sup> Q1 journal rankings sourced from SCImago Journal & Country Rank portal.

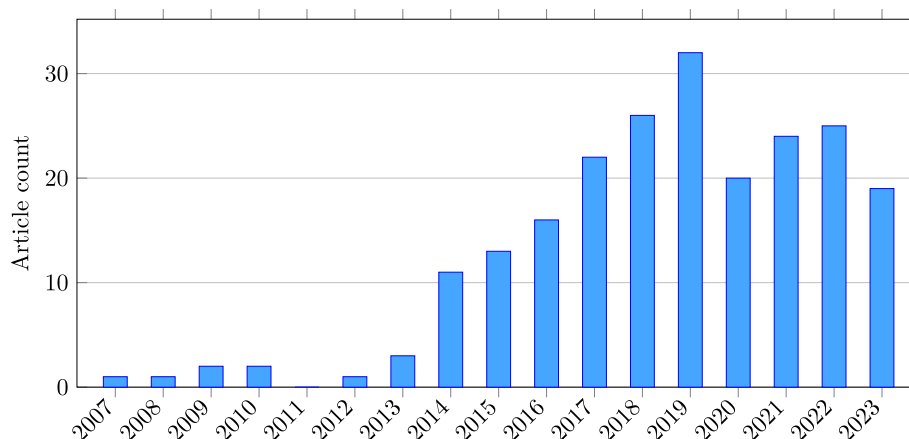


Fig. 4. Articles per publishing year.

Table 8

Most published authors.

| Author                 | Article count | Articles   |
|------------------------|---------------|------------|
| Gütl, Christian        | 6             | [47–52]    |
| Yang, Diyi             | 5             | [53–57]    |
| Alamri, Ahmed          | 4             | [58–61]    |
| Chen, Chen             | 4             | [62–65]    |
| Lepp, Marina           | 4             | [66–69]    |
| Luik, Piret            | 4             | [66–69]    |
| Rosé, Carolyn Penstein | 4             | [53–56]    |
| Sadler, Philip M.      | 4             | [62–65]    |
| Sonnert, Gerhard       | 4             | [62–65]    |
| Vitiello, Massimo      | 4             | [50–52,70] |

### 5.1. Bibliographic analysis

As seen in Fig. 4, the increase in interest in e-learning research has been strong between 2013 and 2019, likely due to the great interest in MOOCs. From then on, the intensity has dropped somewhat.

We analysed the contributions of all authors across the 216 articles (not just the first author). Although most of the authors appeared infrequently, a notable number had published multiple articles on the topic. The authors who had written four or more articles are depicted in Table 8. There were also 18 authors who have written three articles each and 40 who had written two.

As quite a sizeable portion of the articles were from recent years, the citation count is not a precise descriptor of interest in the articles. However, it gives some perspective about the amount of interest the most cited articles have raised. Articles with more than 400 citations are included in supplementary material (see Table 20), allowing the main text to remain concise while providing detailed insights for readers who wish to explore further. The most cited article, authored by Park and Choi [9], examined adult learners in online courses and found that family and organisational support, relevance, and satisfaction significantly affected persistence. In the article by Hone and El Said [71], the authors identified the relevance of the content and the interaction with the instructors as significant factors related to retention. Onah et al. [12] in their analysis found that many so-called “dropouts” still engage with course material in their own way, highlighting the need for more flexibility in MOOC learning methods.

### 5.2. What has been researched

The research question 1 in Section 1 was as follows: “What has been researched regarding dropout in tertiary-level continuing professional development e-learning courses?”. Our systematic search identified 216 articles relevant to dropout in tertiary-level e-learning contexts

typically populated by adult learners. This relevance was ensured both by including all MOOC studies (given their predominantly adult demographic, as discussed in Section 5.5) and by specifically selecting non-MOOC online courses confirmed to involve adult participants (Section 4).

Within this set of 216 articles relevant to adult learners, we identified a smaller subset of ten studies in which the research was explicitly framed around adult learners as the central focus, evident in their research questions, titles or primary analysis objectives. These studies specifically sought to understand dropout dynamics through the lens of adult learning characteristics, constraints, or contexts. This subset can be further distinguished: several studies are explicitly set within a continuing professional development (CPD) context, such as courses for university staff, professional teachers, or on topics like project management, while others examine general adult education in broader or introductory topics.

Most of the adult-focused studies concentrated on analysing how certain factors (known beforehand) affect dropout. Dirzyte et al. [72] found that lower prior knowledge and certain personality traits (notably among women) increased the risk of dropout in what can be considered CPD-level programming courses, while recognition and rewards were linked to persistence, especially for men. Tello [73] demonstrated that instructional interaction, along with positive attitudes towards it, was strongly related to persistence; situational barriers, such as work and family commitments, were the most common reasons for withdrawal. The article by Park and Choi [9], had the dual objective of finding out about factors related to dropout and which of these factors can be used to predict dropout. They found, as discussed above, that factors highly relevant to CPD learners, family and organisational support, relevance, and satisfaction, affected persistence. In a clear CPD context, a study of online university pedagogy courses for university staff by Laato et al. [74] showed that dropout was most likely at the beginning of the course and that adding an initial engagement task improved retention.

Two of the articles were randomised controlled intervention studies. In the study by Azzolini et al. [75], which focused explicitly on CPD, the intervention was conducted in the countries of the European Union and Turkey. The study showed that personalised support, delivered as targeted messages based on participant characteristics and course behaviour, significantly improved completion rates in online professional development courses for teachers in the EU, increasing the completion rate of the course from 32% to 42% among professional teachers. The effect was most pronounced among those with little prior online training experience or those who were slow to start the course. The intervention was cost-effective but had limited impact among student teachers and among all participants in Turkey. In the study by Janssen et al. [43], analysing a general adult education course on using the

Internet, the trial consisted of providing learners with navigational feedback, basically recommendations based on the successful course choices of their predecessors. It clearly increased the number of learners who completed all units of learning, compared to a control group, from 33% to 40%.

Two articles had the aim of building a theory, model, framework, or similar. In the first, by Heilporn and Lakhal [76], the aim was, quoting:

The study aimed at building a reliable and valid scale to study environmental facilitators or barriers related to student persistence in online courses that are particularly suitable for adult or lifelong learner students.

The proposed scale was based on an existing student progress model and was tested with a group of participants in online university courses. The first group of almost 400 persons was used to explore the data set and to uncover valid and reliable environmental and persistence scales. The second group of similar size was used to confirm the validity of the construct. In the second article by Chaker et al. [77], which studied a project management MOOC, a structural equation model (SEM) indicated that intentions related to extrinsic motivation, highly relevant, such as pursuing a certificate, participating in academic research, or having socialising goals, are associated with greater persistence in MOOCs. Regarding significant predictors of reduced dropout, flow via antecedents such as social presence and variables, interactivity, and telepresence [78], was found to be influential.

In the study by Benda et al. [79], interviews were conducted with adult students who were enrolled in introductory computer science online courses, a context more aligned with general adult education than CPD. The study identified that the main contributors to dropout were the unpredictable and demanding nature of programming assignments and the challenges these presented to managing time alongside work and family obligations.

Lastly, prediction of dropout within a CPD context was included in the article by Cobos and Olmos [80] using machine learning techniques.

To summarise, those studies explicitly situated in a CPD context did highlight factors directly tied to career progression and specific professional skills. For example, motivations such as “pursuing a certificate” [77] and “rewards and recognition” (particularly for males) [72] emerged as significant persistence factors. Furthermore, the importance of “organisational support” [9], holds clear relevance for CPD learners. A tendency for dropout to occur in the “early stages” of courses was also noted in a CPD setting [74]. Studies focusing on broader adult education contexts within this subset tended to emphasise more universal adult learner challenges, such as “work and family commitments” [73,79]. General online learning mechanics like “instructional interaction” [73] and “navigational support” [43] were also highlighted.

The remaining 206 articles, while conducted in settings relevant to adult learners (predominantly MOOCs), did not explicitly frame their analysis around the unique dimensions of adult dropout. Instead, their research focus was typically broader, examining dropout factors, prediction models, or interventions within the general context of the online course or MOOC studied, without centring the analysis specifically on “adulthood” as a primary variable. However, these studies provided valuable empirical data on dropout phenomena within environments highly relevant to tertiary-level CPD, contributing to the general map of the research landscape presented in subsequent sections. The scarcity of studies explicitly focusing their research questions and analysis on the adult dimension highlights a gap, suggesting that the specific challenges and motivations of adult learners in CPD e-learning warrant more direct investigation.

### 5.3. Research questions in articles

The research questions and the main topics of the article were classified. If an article had multiple research questions, only the ones about dropout were included in our classification.

The classification categories arose from the actual content of the articles. The categories and their descriptions are depicted in Table 9, and in Fig. 5. Many articles had several classes of research questions, so the sum of the classification results in Fig. 5 is more than the number of articles.

In Fig. 6 a thematic classification of research questions is presented, as used as an example by Thomas and Harden in [111]. The numbers show how many times a certain category of research questions occurred in the core publications (as defined in Section 4.7). The figure serves a dual aim of both showing a higher level of research question categories and also describing the distribution of which questions have received wide coverage in the core literature and which have not.

The data show a clear concentration of research in three areas: *factors and causes* (17 articles), *remedies and interventions* (16 articles), and *prediction* (11 articles). An interesting finding is the imbalance within *factors and causes*, where studies of *certain factors* (15) vastly outnumber research into direct *causes* (2). Regarding predictive approaches, *ML prediction* (10 articles) is by far the most common approach to prediction.

In contrast, other areas of research are far less common. The entire *definitions, descriptions, theories and models* category contains only 5 articles in total. *Literature review* (2 articles) is also infrequent, while *instructional design* and *replication* are entirely absent from the core literature set.

Table 10 contrasts the thematic categories of research questions in the core publications and in the entire set of articles. The table clearly shows a remarkable consistency between the Q1 journals ( $n = 37$ ) and the complete data set ( $n = 216$ ), both showing similar distributions across the categories of research questions. The core publications do have a larger proportion of theory- and definition related articles and subsequently fewer prediction articles than the whole data set. This suggests that the articles in the core publications are inclined to tackle more difficult problems than relatively straightforward prediction.

### 5.4. Research design, theories, and analysis methods

Regarding the **research designs** used in the articles, quantitative techniques were dominant, accounting for 87.5% of all designs. The qualitative portions were thus much less common, with a share of 7.9%, and 4.2% were classified as mixed-method studies. An article has only been categorised as a mixed-methods one if it had been explicitly declared so. If there were just quantitative and qualitative sections, both got a separate count to those categories instead.

The review articles gathered a share of 5.6%. These have already been discussed in the Section 2.

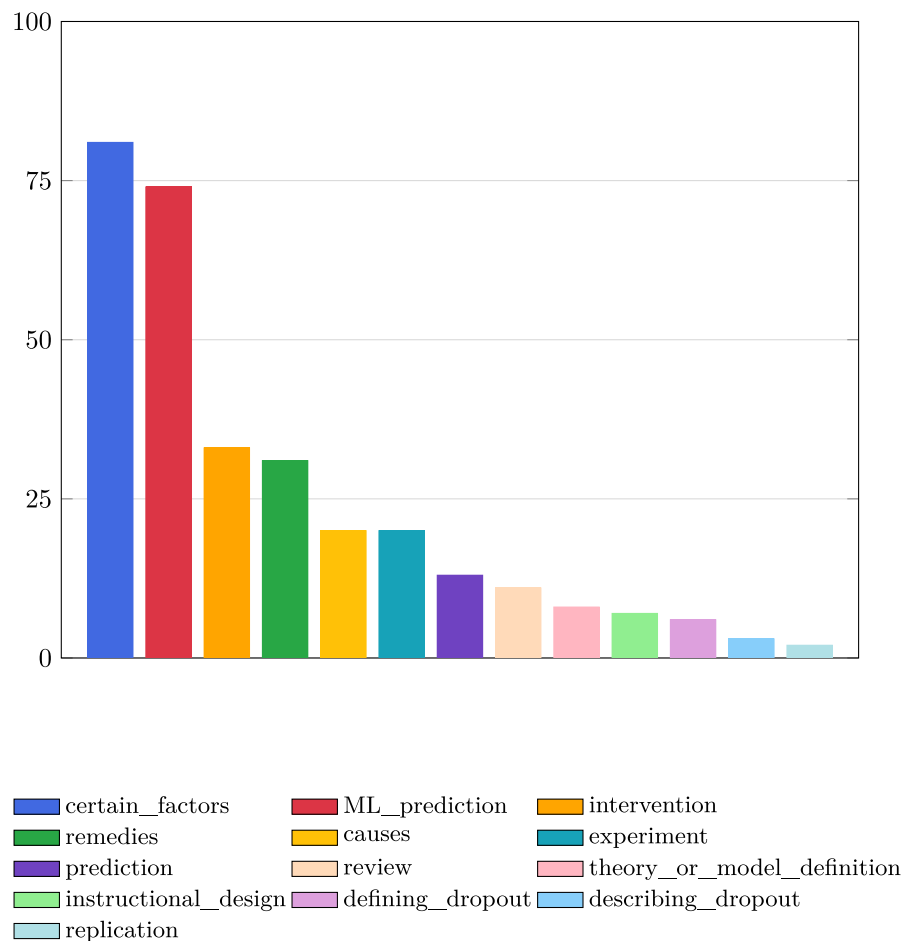
The “not\_applicable” category (0.5%) has only one article, which describes the creation of a system designed to help reduce dropout [112].

With regard to **theories, models, frameworks, and methods**, we identified and analysed all conceptual and methodological approaches related to dropout or persistence presented in the articles. In addition to identifying these approaches, we also analysed how each was used within the reviewed studies, for example, whether a theory was newly created, further developed, or applied to a certain context. Although we distinguished between “theory”, “model”, and “framework”, we recognise that the boundaries between these terms are often ambiguous in the literature. For this reason, we have not included a deeper discussion of the nuances of these terms, as such a categorisation was not relevant to our research questions.

*Influential theories such as Rovai’s sense of community [113] and the composite persistence model [114], as well as Tinto’s model of student*

**Table 9**  
Research question classification.

| Category                   | Description                                                                                                                                                                                          |
|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| certain_factors            | The relation of some factors (decided before the study) to dropout have been investigated. Examples are: [81,82], and [65].                                                                          |
| ML_prediction              | Prediction of dropout using machine learning techniques. Examples: [83,84], and [85].                                                                                                                |
| intervention               | Some kind of intervention was carried out (always implies also the “remedies” category in the classification count); examples are [86–88].                                                           |
| remedies                   | Analysis of factors that helped to mitigate dropout. For examples see [20,89,90].                                                                                                                    |
| causes                     | Causes for dropout, which are unknown in advance. Examples include [91,92], and [93]. It is worth mentioning that it is most often combined with other categories, mostly “certain_factors” [10,94]. |
| experiment                 | A true or quasi-experiment (always implies also the “remedies” and “intervention” categories in the classification count). Examples: [95] (true), [96] (quasi), and [97] (true).                     |
| prediction                 | Prediction of dropout using something else than machine learning techniques, e.g. linear regression. An article having only this category is [55].                                                   |
| review                     | A review. These articles are discussed in Section 2.                                                                                                                                                 |
| theory_or_model_definition | Presenting theory or model for dropout. The five articles having only this category are [76,98–100], and [77].                                                                                       |
| instructional_design       | Dropout-related instructional design for courses. Examples include [42,101], and [102].                                                                                                              |
| defining_dropout           | Dropout definition. Serving as examples, articles having only this category are [103–105], and [13].                                                                                                 |
| describing_dropout         | Describes the dropout phenomenon, for example, as at what point of a course most of dropout happens. The four articles having this category are [106] (has only this category), [68,107], and [108]. |
| replication                | A replication of another study. This category was only found in two articles, [109,110].                                                                                                             |



**Fig. 5.** Research question classification. The total number of values is 309 because some articles provided separate values for multiple categories.



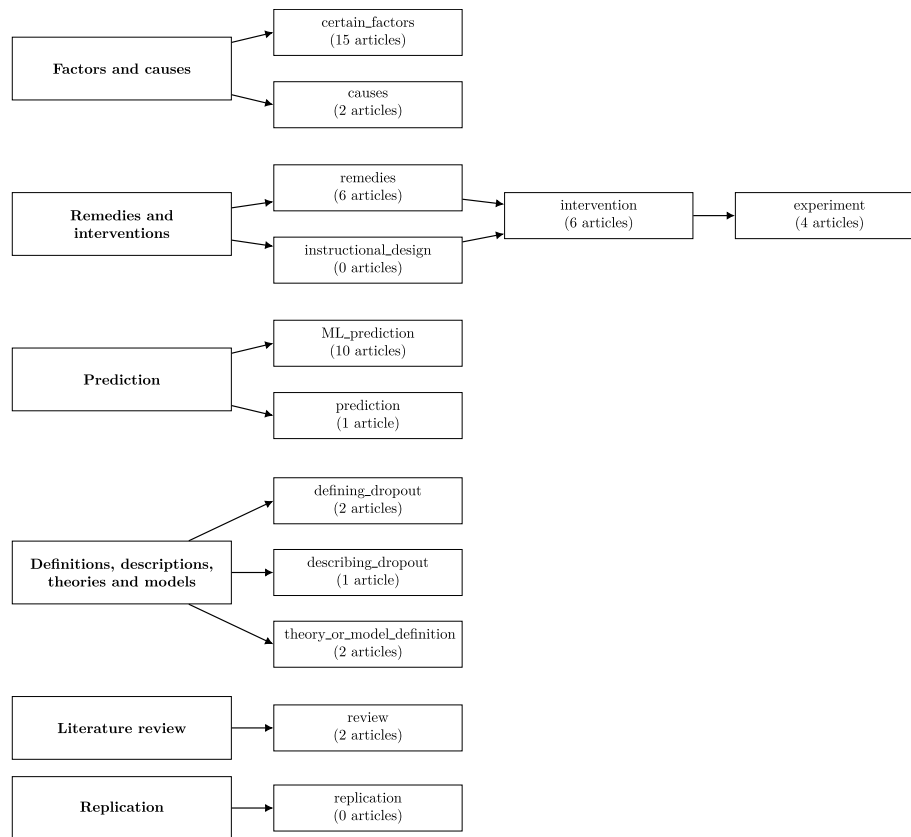


Fig. 6. Classification chart of research questions including counts in core publications.

**Table 10**  
Research question thematic category counts.

| Thematic category                              | Core publications |            | Full article set |                        |
|------------------------------------------------|-------------------|------------|------------------|------------------------|
|                                                | Count             | %          | Count            | %                      |
| Factors and causes                             | 17                | 33         | 101              | 33                     |
| Remedies and interventions                     | 16                | 31         | 91               | 29                     |
| Prediction                                     | 11                | 22         | 87               | 28                     |
| Definitions, descriptions, theories and models | 5                 | 10         | 17               | 6                      |
| Literature review                              | 2                 | 4          | 11               | 4                      |
| Replication                                    | 0                 | 0          | 2                | 1                      |
| <b>Sum</b>                                     | <b>51</b>         | <b>100</b> | <b>309</b>       | <b>101<sup>a</sup></b> |

<sup>a</sup> Total percentage exceeds 100 due to rounding.

*attrition [5] were excluded from our main analysis, as they do not directly address adult online learners. Their influence is discussed where relevant.*

One of the most important findings of this review is the scarcity of explicit theory use in the articles. Among the studies that did use a theory in dropout or persistence research, there was considerable variation, with examples such as self-regulated learning (SRL) [82, 115–117], expectancy-value theory [63,65], self-efficacy theory, goal theory [118], and the reasoned action approach (RAA) [105]. Although these theories are widely recognised in educational research, their direct use to explain or predict dropout was limited to only eight studies.

Some articles proposed or applied frameworks specifically to address dropout and retention. Hall's high-/low-context culture framework was used to interpret dropout patterns between learner groups [119]. Two studies created new frameworks for this purpose, including one for designing dropout reduction interventions [120] and a strategic framework for evaluating retention in MOOCs [121].

Five studies also introduced or applied methodological approaches focused on dropout research. These include the use of the DEMATEL<sup>6</sup> method to map the relationships between dropout factors [122,123]. The grounded theory method (GTM) was used to generate a new retention model [124], and thematic analysis was used to explore participants' perspectives on success and completion in MOOCs [104]. Phenomenological analysis was used to create a description of the reasons for dropping out [79].

We did also categorise how the models and methods were *used*; the results are listed below.

**Creation** A new theory, model, framework, or method was created, for example, using the Grounded Theory Method (GTM). In addition, cases where a new model has been created by combining two or more other theories (or models or similar), and then given probability coefficients based on the data set, have been counted in this category. In total, 12 articles belong to this category. Seven of these consisted of the creation of a structural equation model (SEM), which is discussed below. As examples of creation, GTM was used by Adamopoulos [124], a model based on the flow concept has been developed by Chaker et al. [77], and Pant et al. [121] created a strategic framework for evaluating and responding to the causes of retention for MOOCs.

**Furthering** A theory or model has been created based on existing ones, or it has been somehow refined. Examples include [99,115]. Together, there were five articles with this classification.

**Application** A theory, model, framework, or method was used in the interpretation of the results of an article. Examples: [96,117, 119]. This category contained 12 papers.

<sup>6</sup> Decision-Making Trial and Evaluation Laboratory.

The *models* used in the various theories can be divided into three categories, which are described in detail in the supplementary material (see [Appendix A.2](#)).

As a special note, we did not classify the mathematical approaches used in articles related to dropout prediction — these have been dealt with in earlier reviews [15–17,125].

In the end, only 28 papers, or 13% of our whole data set of 216 articles, had used some kind of theoretical approach. By omitting the 66 articles having machine learning prediction or other prediction as their only research target, the percentage increases to 18.7%.

Regarding **analysis methods**, among *quantitative* approaches, advanced statistics (such as regression or significance testing) were the most common (94 articles, 39.8%), followed by machine learning techniques for prediction (72 articles, 30.5%), and descriptive statistics (for example, reporting percentages) (42 articles, 17.8%). For *qualitative* analysis, 13 articles (5.5%) used a declared qualitative method (such as thematic or phenomenological analysis), while 10 articles (4.2%) simply presented interview samples or limited qualitative content without a deeper analysis. Five articles (2.1%) were classified as not applicable. It should be noted that a single article may have used several methods and, as such, the total count exceeds the number of articles; a more thorough description of the analysis methods, including examples, is included in the supplementary material ([Appendix A.3](#)).

## 5.5. Data sources

### 5.5.1. Overview of the whole data set

Regarding **research setup**, some articles are based on courses of which the researchers can themselves decide what features they want to extract, and some are based on pre-existing datasets, or on participant surveys.

The most common were direct course data (59 articles, 27.3% of all the 216 articles) and interview or questionnaire responses (55, 25.5%), followed by other non-public datasets (47, 21.8%) and public datasets (39, 18.1%). Articles frequently used more than one data source, so the percentages above sum to more than 100%.

Fewer articles were reviews (12, 5.6%). Some relied on unclear or unspecified sources (14, 6.5%) or were classified as not applicable (4, 1.9%).

Longer descriptions and examples of each category are provided in the supplementary material in [Appendix A.4.1](#).

**Course types:** almost all articles concentrated on MOOCs (total count 195, or 90.3%); online courses were much less common (22/10.2%). One review article was targeting both MOOCs and online courses [15]; it has been included in both counts above.

Various **kinds of data** were used to analyse dropout and persistence. Again, there can be more than one data source in an article, so the following percentages sum to more than 100%. The most common were recorded activities (for example, participation in quizzes or forums; 126 articles, 58.3% of all the 216 articles), and clickstream data (low-level logs of student actions such as keystrokes or mouse clicks; 112, 51.9%). Questionnaire data (fixed or open-ended survey responses) was also frequent (70, 32.4%), as were course results (grades from examinations, homework, or assignments; 60, 27.8%). Demographic information was either analysed in relation to dropout (44, 20.4%) or only used to describe the dataset (28, 13.0%). Only 10 or 4.6% articles included in-person interviews.

Some articles relied on reviews, theory-building, or simulations (13, 6.0%), and a few used other data types such as student feedback or computer-generated data (7, 3.2%).

Further descriptions and detailed examples for each category are provided in the supplementary material in [Appendix A.4.2](#).

The **course subjects** were classified according to the *ISCED categorisation of the fields of education and training* [46], applied only to courses not based on public datasets. The most frequent categories were ICT (71 courses, or 32.9% of all the 216 articles), business/administration/law

(35, 16.2%), STEM (34, 15.7%), and arts/humanities/languages (30, 13.9%), with clearly fewer articles in other subject areas. Full details with example course titles and exact counts are in the supplements in [Appendix A.5](#).

Initially, we tried to categorise the **educational levels** of the courses according to the *ISCED classification of educational levels* [27]. Although some articles had quite clearly described the course contents so that the ISCED level was easy to obtain, most only gave an overview. In the end, we decided to use two categories for MOOCs, namely “university-based MOOCs” (46 articles) and “non-formal education MOOCs” (16). For regular online courses, the level categories were university (9) and non-formal education (6).

There may be different design principles in MOOCs depending on the main target audience. They may have been designed from the ground up as open courses (“conventional MOOC”, as defined in [126]), or they may be university MOOCs designed for lifelong learning (“professional MOOC”). A MOOC can also be a normal university course that is offered as a MOOC (“formal MOOC”). However, we decided to use the categorisation given in the previous paragraph, as this information was very difficult to obtain from the source articles.

Course **workloads** turned out to be a rare figure to give. Only one article had given the workload in the ECTS system as one credit [66], and eleven more had estimated the workload as an exact time amount. These exact amounts for two articles were rather small, three and four hours [127,128]. The remaining nine articles had the amount in dozens of hours.

Course **durations**, on the other hand, were quite often given for MOOCs. They were typically either short (1–9 weeks) or semester-long, as detailed in the MOOC characteristics in Section 5.5.2. The **numbers of participants** in the courses naturally varied a lot. These have been depicted in [Fig. 7](#), which lists the count of articles with a certain number of students.

### 5.5.2. Characteristics of the MOOC studies

For **MOOCs**, we also analysed the aspects listed below. The additional 22 online courses included in this review were all specifically selected for their focus on adult education, making a further classification of their “adulthood” unnecessary.

To address the generalisability of MOOCs to the broader CPD landscape, we categorised them according to the delivery platform, the type of provider, and the regional context (see [Table 11](#)). The majority of studies in which the platform was specified used global commercial platforms (82 studies), such as Coursera or edX. National or government-led platforms (e.g., XuetangX) accounted for 33 studies, while 16 studies examined courses hosted directly on university platforms (for example, Moodle). Geographically, the literature is dominated by providers in Europe (55) and North America (36), followed by East Asia (29). Fewer studies originated in Latin America (13), the Middle East & Africa (4), and South Asia (3).

With regard to demographic data, it is important to note that the percentage of people who give such information can be very small. For example, in [129] only a bit more than 10% of the 284 888 students had submitted their age or educational background. This problem is also discussed in [130]. In the European Union, GDPR legislation [131] can even forbid asking for personal information unless for research purposes.

**Taxonomy of learning objectives** None of the MOOCs directly used an actual taxonomy of learning objectives [132,133] to describe intended knowledge levels. Only eighteen articles included any description of course aims. We chose not to classify courses by learning objectives, as fewer than 10% provided such descriptions and trying to find taxonomy levels from brief statements would be unreliable.

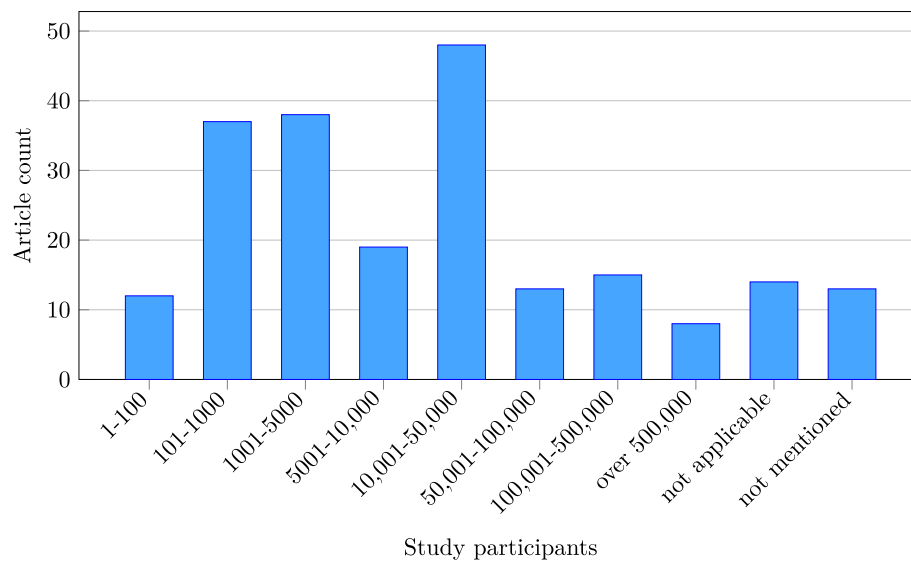


Fig. 7. Articles per participant count.

**Table 11**  
Contextual characteristics of the included MOOC studies.

| Category                | Sub-category                  | Count |
|-------------------------|-------------------------------|-------|
| Delivery platform       | Global commercial             | 82    |
|                         | National/government           | 33    |
|                         | University hosted             | 16    |
|                         | Domain specific               | 7     |
|                         | Not mentioned                 | 65    |
| Content provider type   | Higher education institution  | 121   |
|                         | Commercial                    | 8     |
|                         | Government/public institution | 2     |
|                         | Not mentioned                 | 65    |
| Content provider region | Europe                        | 55    |
|                         | North America                 | 36    |
|                         | East Asia                     | 29    |
|                         | Latin America                 | 13    |
|                         | Middle East & Africa          | 4     |
|                         | South Asia                    | 3     |
|                         | Oceania                       | 1     |
|                         | Not mentioned                 | 56    |
| Course language         | English                       | 74    |
|                         | Chinese                       | 23    |
|                         | Spanish                       | 12    |
|                         | German                        | 5     |
|                         | Estonian                      | 4     |
|                         | French                        | 4     |
|                         | Others                        | 5     |
|                         | Not mentioned                 | 74    |
| Participation scope     | National                      | 26    |
|                         | Global                        | 13    |
|                         | Regional                      | 8     |
|                         | Not mentioned                 | 149   |

**Prerequisite information** Prerequisite information for MOOCs was rarely described in the articles, only nine mentioned any. Two specified prerequisites such as compulsory schooling, university-level physics, or “professional knowledge”, while seven articles explicitly stated that there were no prior requirements (e.g. [68, 119]).

**Target group** The MOOC’s intended audience was described in 22 articles — more frequently than prerequisites. Thirteen stated the course was designed for everyone (examples: [87,134,135]),

and six targeted specific professionals, mostly teachers (examples: [136,137]). Of the rest, one MOOC was aimed at everyone but with an age limit of 16+ [103]; one was aimed at students [120], and one study covered several MOOCs with different target groups.

**Course duration** Course durations were often specified. They were mostly between one and nine weeks (65 articles) or a full semester (43). Four studies did not have a duration limit [117, 138–140]. However, it is important to note that our review did not classify courses as scheduled (instructor-paced) versus self-paced, which can affect the interpretation of duration. In self-paced MOOCs, the listed duration is often administrative rather than a reflection of an actual learning period.

**Participant age** Older adults typically made up the majority of MOOC participants (see also [141, p. 4]). For example, Kizilcec and Halawa [142] reported a weighted average age of 33.6 years among 445 000 participants. Reporting methods varied between articles, with most using either age categories or averages. Calculating a weighted average of the latter from 31 articles, which had reported an age value, gave an overall mean value of 32.0 years.

**Participant employment status and professional field** Only fourteen articles reported the employment percentage, either describing populations of employed individuals or mentioning employment status as part of participant characteristics (for example, [67,71,86,93,127,143,144]). The reported values varied considerably, ranging from 21.0% to 100%, with a median of 59.3%. In the majority of these cases (9 of 14), the employment rate exceeded 50%.

Specific details on the professional field of the participants were even scarcer, appearing in twelve studies. In these cases, the reported backgrounds were predominantly in *education* (e.g., school teachers, university educators) [74,101,126,136, 137,143] and *STEM* sectors (e.g., engineers, IT specialists) [10, 42,145]. Other profiles mentioned included management roles (team leaders, project managers) [146], specific industry employees (food manufacturing) [127], or a mixed cohort consisting of professionals alongside university students and general graduates [147].

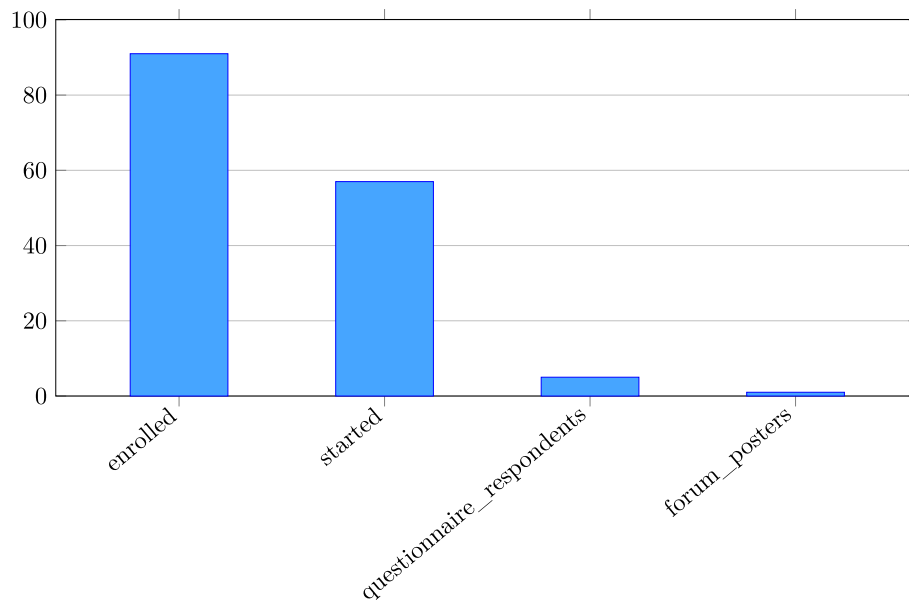


Fig. 8. Dropout definition source. Data was given in 142 articles. The total number of values is 154 because some articles provided separate values for multiple categories.

**Participant educational level** Thirty-six MOOC articles reported participant education in their demographics. We treated bachelor's degree as the baseline, as it is a common final degree in the U.S. [148]. Most studies found that MOOC participants tend to have high education levels [104,149], though courses specifically aimed at undergraduates or students (or when students were respondents) had few postgraduates [73,82,150,151]. In some cases, the university degree and the employment status were mutually exclusive, lowering the degree level [144,152]. Some courses targeted degree holders or master's students ([10], 85% bachelor or higher), and non-random samples sometimes yielded very high rates of degree holders ([104], 90.9%). In total, 20 articles reported that more than 50% of participants held at least a bachelor's degree. An additional four articles found that 35–50% held a bachelor's degree, while eight articles reported 20% or fewer with this qualification.

### 5.6. Dropout definition

Dropout is a very complex phenomenon, affected by multiple aspects and points of view. First, there is a temporal dimension of dropout ([153, p. 298] [23, p. 3]). For example, a student may enrol in a course but never even enter the course system, or dropout may happen later during the actual course.

Regarding this temporal dimension, the articles were classified according to the number from which the dropout percentage was calculated. The most typical source number was the number of enrolments (also called registrations in some sources), quite a few also used the number of students who actually started the course. Some articles (for example, [81,149,154]) have used both figures separately. In one case, there were two sources of courses, where one set of data only included participants who had started a course, and the other contained all the enrolments [70]. The results of this classification are shown in Fig. 8.

Continuing with the temporal aspect, the next classification concerns what the criteria have been for being considered a pass.

Passing a course may have several definitions. Fig. 9 shows the relationships between dropout, failure, completion, and passing a final exam. In quite many cases, failure has been the definition of dropout, although a person may have completed all the material and assignments (for a discussion, see [79, p. 2]). It is important to note that the numbers in the picture denote the number of articles having a certain definition.

In some cases, an article had defined and analysed dropout and failure separately, and such an article has been counted in both categories.

In relation to the definition of dropout as given in the articles, the categories of passing a course or being a dropout are shown in Figs. 10 and 11, respectively.

The percentage of dropouts is an interesting result of any dropout-related study. To give a visual description of how the values are distributed, dropout percentages have been categorised as 0%–20%, 20%–40% and so on. As discussed earlier, visualisations have been separated between enrolments and actual starters. MOOCs have been treated separately as there has been a lot of discussion about the high numbers of dropouts (for a review, see [23]).

It is interesting to see that there is a difference in the dropout percentage profiles between online courses and MOOCs when the values have been calculated from the time of enrolment. For online courses, the rates were generally lower ( $M = 44.6\%$ ,  $Mdn = 37.4\%$ ,  $SD = 21.1\%$ ,  $n = 8$ ). As seen in Fig. 12, the dropout percentages for MOOCs are in this case high ( $M = 77.8\%$ ,  $Mdn = 85.5\%$ ,  $SD = 22.7\%$ ,  $n = 56$ ), with a full range spanning from 4% to 100%. It should be noted that these figures represent an unweighted average of the reported rates; a participant-weighted calculation was not feasible, as only 12 studies provided both the dropout rate and the participant count.

As the figure reveals, when the percentages have been calculated from the time the course content was actually started, the average MOOC dropout rate is marginally lower ( $M = 72.6\%$ ,  $Mdn = 82.0\%$ ,  $SD = 21.5\%$ ,  $n = 36$ ), although the large standard deviations and the different samples involved caution against drawing strong conclusions from this comparison alone. For online courses, the average dropout rate calculated from the start was 50.6% ( $Mdn = 55.0\%$ ,  $SD = 19.2\%$ ,  $n = 5$ ). Several articles argue that “started” is conceptually the more appropriate baseline for calculating dropout, as it excludes individuals who enrolled but never engaged with the course [70,103,156]. The problem of inflated rates due to no-shows is also highlighted in [154]. Given the variability in our findings, future studies would benefit from systematically comparing dropout rates using both calculation methods within the same course to better understand the practical difference between these approaches.

Continuing on the dropout rate problems in MOOCs, it has been pointed out that in many cases students are not even interested in “completion” in the traditional sense [10,13,52,147,157]. Instead, they simply aim to fill in any gaps in their knowledge, and are quite happy

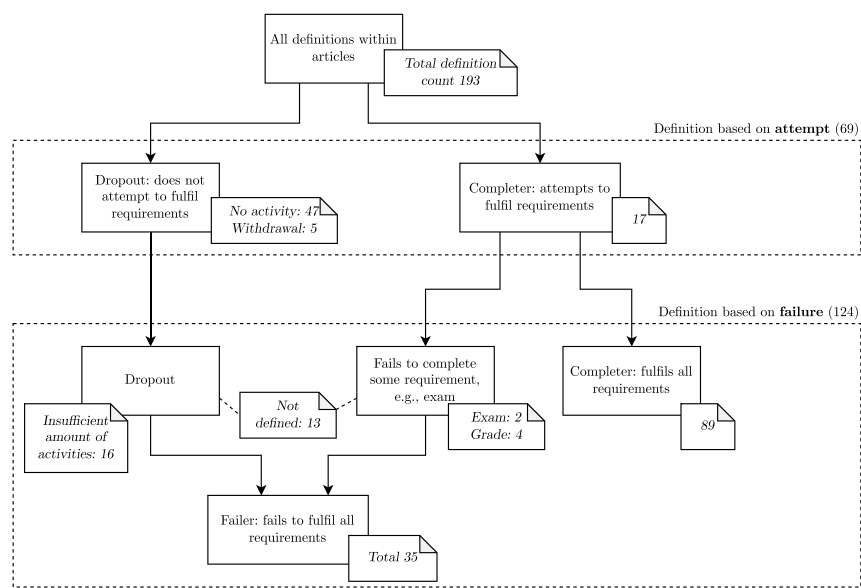


Fig. 9. Dropout definitions, numbers representing count of articles with a certain definition. Source: Adapted from [155, p. 89].

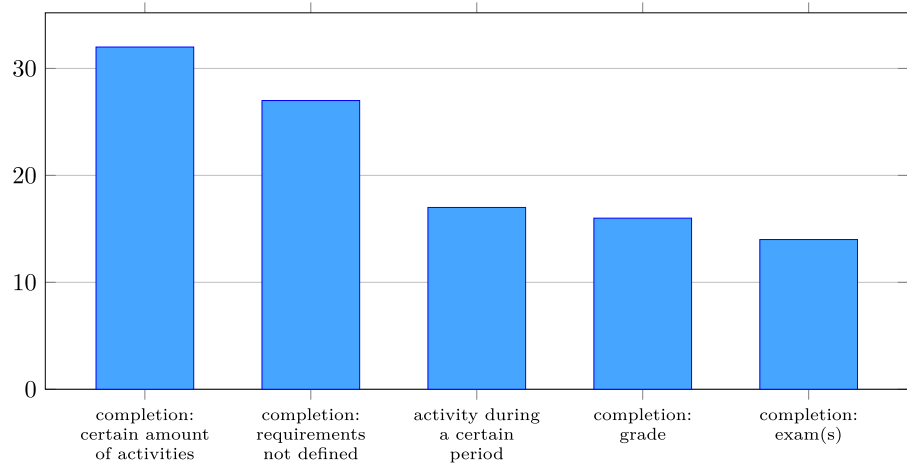


Fig. 10. Criteria for passing. Data was given in 100 articles. The total number of values is 106 because some articles provided separate values for multiple categories.

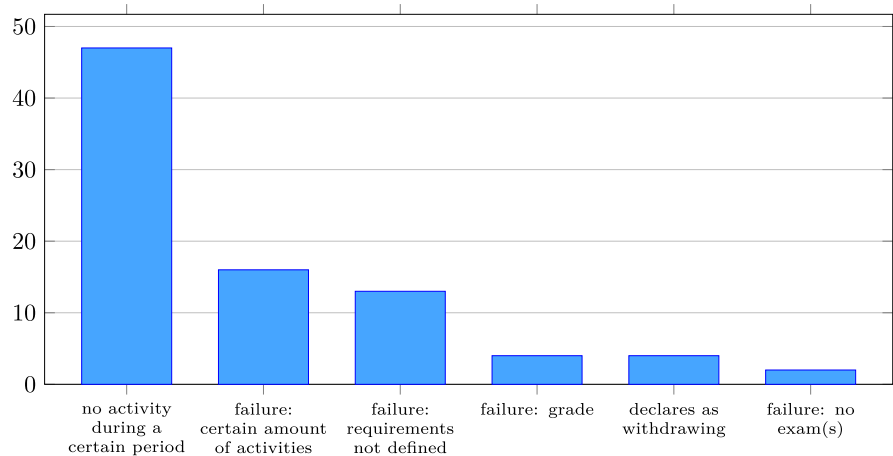


Fig. 11. Criteria for dropout. Data was given in 80 articles. The total number of values is 86 because some articles provided separate values for multiple categories.



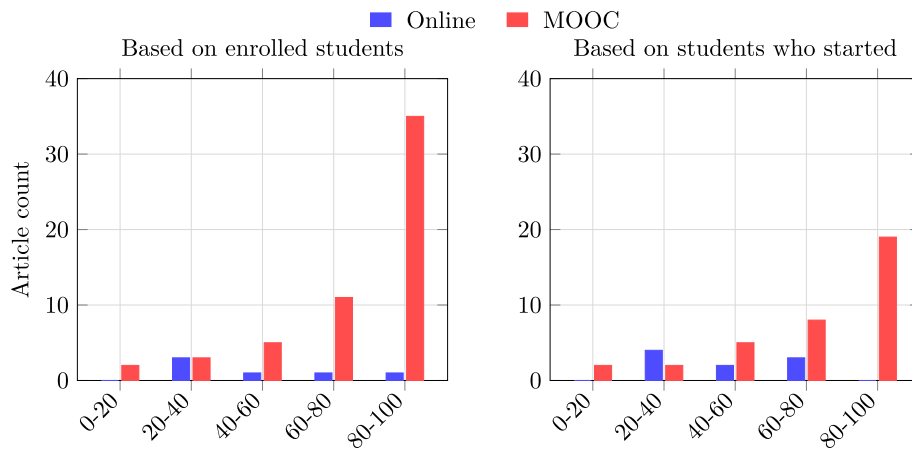


Fig. 12. Dropout percentages, from *enrolling on* or *starting* a course.

when that goal has been reached. This aspect further gives worse completion figures for MOOCs compared to online courses, unless it is taken into account when amounts are being calculated. In a similar sense, abnormal learners who skip videos but still get good grades can make dropout prediction more difficult [158].

These findings would indicate that it is likely that the way MOOCs are used is behind the high dropout rate, instead of MOOCs being bad as a type of online courses. In fact, the case would be the opposite if MOOCs can gather such an audience that would otherwise skip learning altogether [159].

Regarding the **viewpoints and learner typologies** related to dropout definitions, Whitehill et al. [97] review several common definitions of dropout based on an *external*, observable viewpoint:

- The last event performed by the participant; submitting some kind of assignment, watching content, posting to a forum, or basically any such event
- Certificate or examination
- Hybrid definition, such as having watched fewer than 50% of the course's videos and having executed no action during a certain period

Another article goes a long way to providing even a formula for counting how long a break a student can take [14]. Although that definition is sophisticated in its criteria, it is still only observational, as it does not capture internal student motivations or experiences.

Several studies identify additional categories for participation [13, 103, 160, 161]. These classifications often highlight the presence of casual learners, who pick and choose only the content most relevant to them [104, 152, 156, 157]. The resulting participant types are summarised as follows:

1. No-shows: The student registers for the course but never logs in to the course to interact with the content
2. Observers/samplers: Wants to see what an online course is like or how this one is taught
3. Casual learners: Wants to learn one or two new things, either out of curiosity or a work/school-related need. These can further be divided into:
  - Auditing: Studies most of the content but completed no or only some activities
  - Disengaging: Completes the activities at the start of the course and then reduces their engagement
4. Completers: Finalises as many course elements as necessary for completion

In [105], success is further divided into two categories, “inclined actors” who fully achieved their individual set of intended goals, and “disinclined actors” who ended up achieving even more than they originally intended.

The largest number of categories in our set of articles is offered by [103] as results of cluster analysis: samplers, strong starters, returners, midway dropouts, nearly there, late completers, and keen completers. Of these, only late and keen completers were not dropouts, and of course it is up to interpretation whether “samplers” are counted as such.

Diverse engagement patterns, a temporal aspect, and high-level categories for dropout and persistence factors are visually conceptualised in *Attrition model for open learning environment setting* (AMOES) [153, p. 298], shown in Fig. 13. This model illustrates the progression from initial awareness through registration and activity to eventual success (the “funnel of involvement”), mirroring the high level of dropout at each step often observed [159, p. 2]. It distinguishes between “persistence”, “unhealthy attrition”, and “healthy attrition”. Within the “healthy attrition” group, the model categorises learners based on their engagement and intentions: “exploring users” wanted a preview of the course or were eager to experience a new learning environment; “content learners” were selective, choosing specific topics of interest without intending to complete the entire course, often due to free access; and “restricted learners” chose to audit the entire course without intending formal completion or participation in assignments or badges.

The distinction made in the AMOES model between healthy and unhealthy attrition aligns with the concepts of voluntary versus involuntary departure also discussed in the literature [52, 79, 147, 157, 162]. In these, voluntary departure is described as the student leaving a course for positive reasons. Examples of such include that they already got the benefits they were looking for, they realised a course is not appropriate for them, they transfer to another similar course, or they have gained employment. So, it is suggested that the word “dropout” is only used in involuntary departure; in cases where implied failure is present. This can be on behalf of either the institution or the individual.

Yet another view of the definition of success is offered by [104], where it was found that the participants in the MOOCs wanted to feel a sense of “closure”. This had varying definitions between participants: application of knowledge (two participants), a sense of closure (one), reflection on own learning (one), and obtaining a certificate (two).

### 5.7. Dropout and persistence factors

The reviewed articles indicate that dropout and persistence in tertiary-level e-learning are associated with a broad set of factors, which are often reported to occur together. Although studies do not typically make strong claims about causality [163], certain themes are

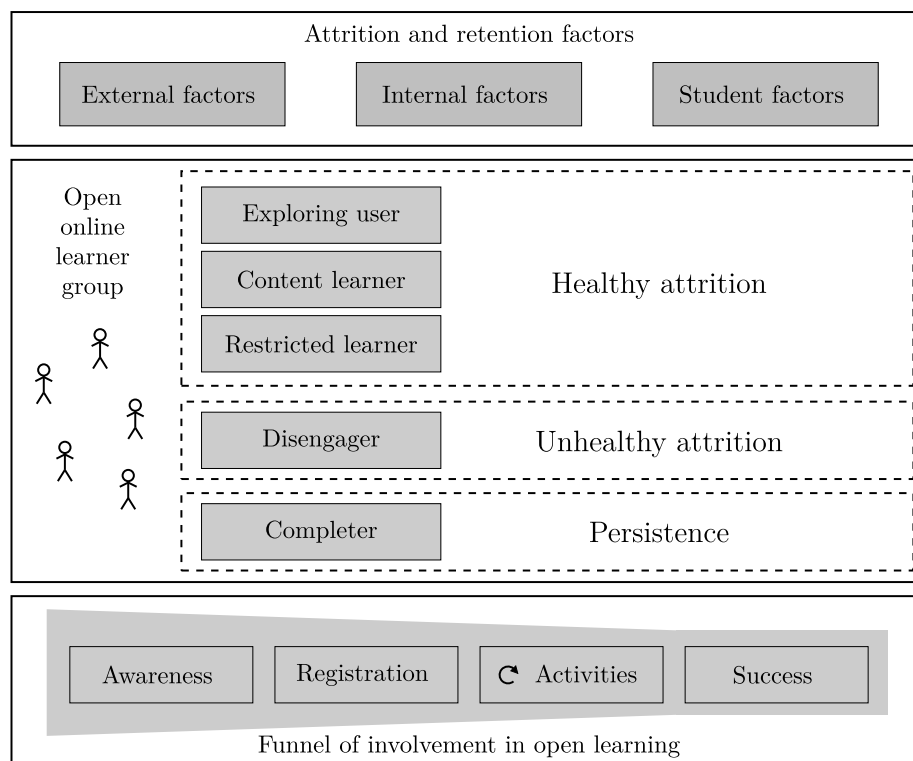


Fig. 13. Attrition model for open learning environment setting (AMOES).

Source: Redrawn from [153, p. 298].

repeatedly identified in different contexts. It is important to note that the factor counts presented below reflect how frequently each category was identified in the articles, not necessarily the weighted importance of that category. Furthermore, to avoid secondary analysis, reasons for dropout were not classified from review articles. So, the following discussion provides a picture of the key challenges and enablers as reported in the primary literature.

A high-level overview of these factors, along with their corresponding pedagogical implications, is presented in Table 12. This view organises the findings according to four overarching themes: individual factors (relating to learner characteristics and constraints), pedagogical factors (concerning course content and instructional design), technical factors (addressing platform and environment issues), and institutional factors (covering administrative and policy aspects). The subsequent subsections present the detailed evidence behind this overview.

Regarding **factors associated with dropout**, our analysis of articles ( $n = 69$ ) that identified specific reasons for dropout reveals a clear hierarchy of challenges faced by learners, as summarised in Table 13 and visualised in Fig. 14. The most frequently cited category is *time*, accounting for a substantial 33.8% of the 142 factors reported. This finding underscores that the difficulty of balancing studies with work, family, and other commitments is the primary reason learners abandon courses.

The next most common category is *motivation and attitude* at 18.3%. This broad category covers issues ranging from a lack of perceived relevance and course satisfaction to misaligned learning styles and negative prior experiences with e-learning. Time-related challenges and insufficient motivation are highlighted in the studies and often appear to reinforce each other [8].

Other significant contributors to dropout include *course content and organisation* (12%) and a learner's *prior capabilities* (11.3%). Issues with the course itself, such as excessive workload, high difficulty, or poor-quality content, are significant contributors to dropout. Similarly, a student's lack of prerequisite subject knowledge, insufficient technical

skills, or language barriers can create obstacles, which are very difficult to overcome.

These findings suggest that, while internal motivation is important, the sheer practical challenge of finding sufficient time is the single greatest predictor of dropout reported in the literature. In contrast, the same two categories (positive versions, that is, enough time and sufficient motivation) are among the most frequently reported factors linked to learner persistence.

In the analysis of **persistence factors** reported in 110 articles, a somewhat more balanced picture emerges between internal and external factors (Table 14 and Fig. 15). Here, *motivation and attitude* is the leading category at 25.7% (calculated from the total of 210 factors reported), highlighting the central role of intrinsic and extrinsic drivers in sustaining learner engagement, such as the relevance of a course in relation to career goals, the availability of certification and encouragement.

*Time* is the second most dominant factor for persistence, at 22.4%. This demonstrates that having sufficient time is not merely the absence of a barrier but a powerful enabler of success. *Course content and organisation* remains a key factor (14.8%), with learners persisting in courses that feature high-quality content and manageable workloads. This is followed by the quality of the *e-learning environment* (11.4%), which includes aspects like system usability and quality of support and feedback.

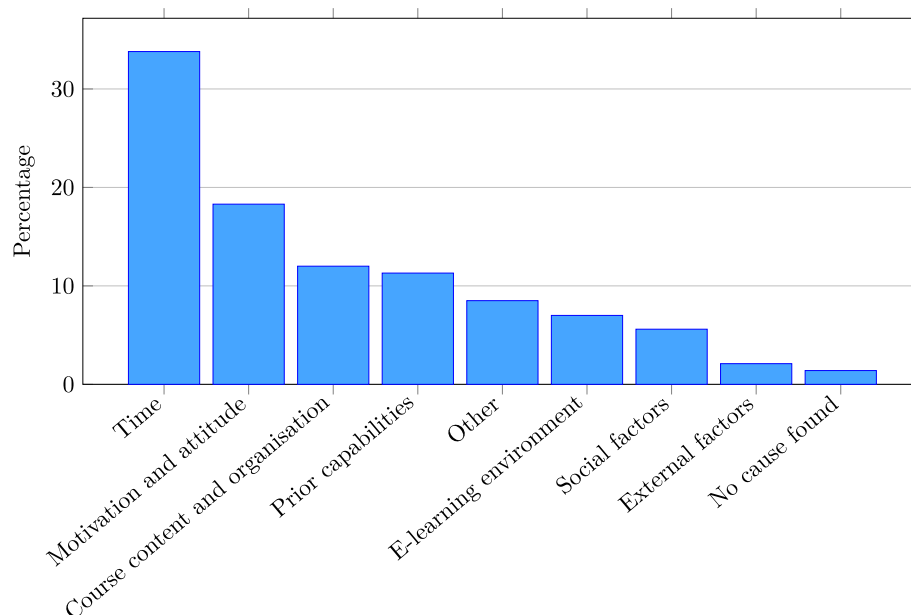
A direct **comparison of dropout and persistence factors** reveals the important role of time. It stands as the primary reason for failure and the second-most enabler of success, just behind motivation. This dual role suggests that effective time management and flexible course design are critical to retaining learners. Although lack of time is the biggest obstacle, motivation appears to be the most powerful proactive driver to completion. Furthermore, some factors, such as gamification and peer feedback, were reported more frequently as supporting persistence than as direct aids against dropout, suggesting that not all persistence factors simply represent the inverse of dropout reasons.

**Table 12**

Synthesis of dropout and persistence factors, effective interventions, and pedagogical implications.

| Theme                 | Key dropout factors                                                                                                                                                                                                             | Key persistence factors                                                                                                                                                                                                                                                       | Effective interventions <sup>a</sup>                                                                                                                                                                                                                                                                                | Key pedagogical implications                                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Individual factors    | <b>Time constraints:</b> The most cited barrier (33.8%)<br><b>Motivation:</b> Lack of relevance, satisfaction, or internal drive<br><b>Prior capabilities:</b> Insufficient subject knowledge, or e-learning or language skills | <b>Motivation:</b> The leading driver (25.7%), driven especially by relevance and learning style match<br><b>Time availability:</b> Having sufficient time is an important enabler<br><b>Prior capabilities:</b> Sufficient subject knowledge, e-learning and language skills | <b>Encouragement:</b> Email nudging, timely encouragement (57 pp effect)<br><b>Rewards:</b> Rewards, especially tangible (47 pp), also job-related commendation (7 pp) or extra privileges in the e-learning environment (6 pp)                                                                                     | <b>Flexible design:</b> Self-regulation support to assist with time constraints<br><b>Motivational features:</b> Use of gamification/other incentives as extrinsic motivators                                                                                                                                                                                                                                      |
| Pedagogical factors   | <b>Course issues:</b> Excessive workload, high difficulty, or poor content quality<br><b>Instructional design:</b> Lack of social features or collaboration                                                                     | <b>Content quality:</b> High-quality material with a suitable workload and duration<br><b>Social factors:</b> Interaction with others, collaboration                                                                                                                          | <b>Communication:</b> Clear course communication and guidance (30 pp)<br><b>Assessment:</b> Pre- and post tests (21 pp)<br><b>Social factors:</b> Human face (8 pp)<br><b>Gamification:</b> Monetary reward, leaderboards, badges (mixed results)<br><b>Adaptivity:</b> Personalised learning paths (mixed results) | <b>Engagement monitoring:</b> Identify disengagement early and react accordingly<br><b>Social learning:</b> Peer teaching, collaborative learning, and other social learning features<br><b>Assessment strategy:</b> Use quizzes for feedback; allow retries<br><b>Structure of progression:</b> Structured progression or adaptivity<br><b>Scheduling:</b> Fixed or open schedule depending on topic and audience |
| Technical factors     | <b>Platform issues:</b> Technical difficulties, poor usability, lack of support features                                                                                                                                        | <b>Usability:</b> Quality of the e-learning platform<br><b>System support:</b> Help and guidance features                                                                                                                                                                     | <b>Feedback:</b> Immediate feedback (15 pp)<br><b>Technical difficulties:</b> Removal of technical difficulties (8 pp)                                                                                                                                                                                              | <b>Interaction design:</b> Facilitation of student–student communication                                                                                                                                                                                                                                                                                                                                           |
| Institutional factors | <b>Cost:</b> Financial barriers (though typically free in MOOCs)<br><b>Credit mobility:</b> No possibility of transferring recognition                                                                                          | <b>Credit mobility:</b> Ability to transfer recognition to, e.g., a degree or work<br><b>Certification:</b> Value of the certificate in employment                                                                                                                            | <b>Rewards:</b> Job-related incentives (e.g., training, CEO recognition) (up to 47 pp effect)<br><i>(Research gap: few direct policy interventions tested)</i>                                                                                                                                                      | <b>Credit mobility:</b> Recognise credits for degrees/work<br><b>Enrolment flexibility:</b> Easy access and flexible entry/exit                                                                                                                                                                                                                                                                                    |

<sup>a</sup> The effect is reported as the arithmetic difference between two percentages (percentage points, pp); for example, a dropout reduction from 40% to 30% is a 10 percentage point (pp) effect.

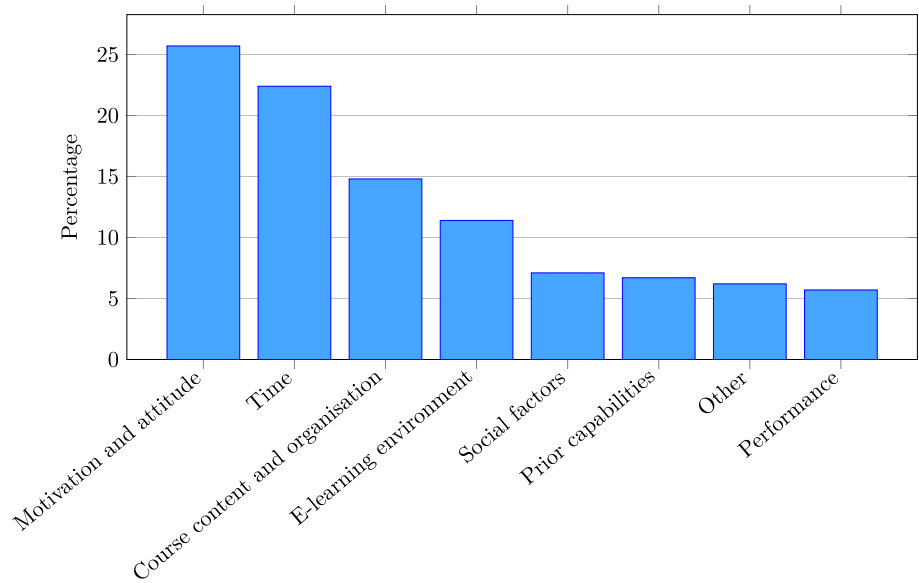


**Fig. 14.** Dropout factors. Data was given in 69 articles. The total number of values is 142 because some articles provided separate values for multiple categories.

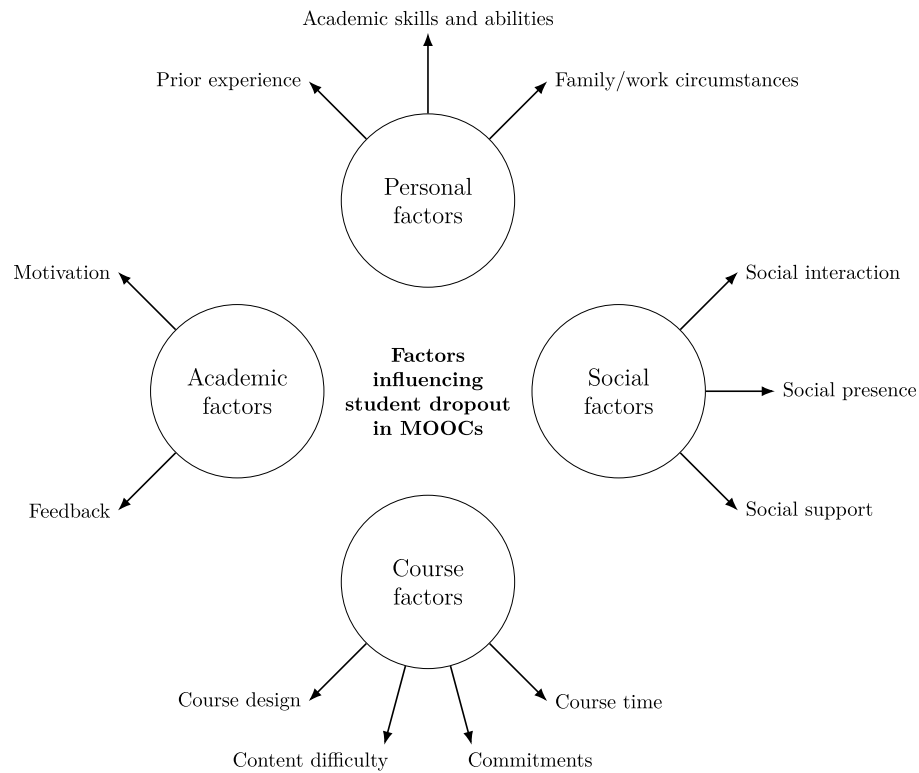
Regarding the **categorisation of dropout factors** found in the set of articles, a relatively comprehensive example of given in [122] is shown in Fig. 16.

Some other recent categorisations in the articles were simpler. In [16] there were two main categories (student and course) and

in [164] three (student, course, and environmental). In [8], quite an extensive analysis with the same three main categories is presented with two levels of subcategories (further analysed in [165]). If we add instructor as a main category based on the findings in [71], the following five categories can be distinguished: (1) student factors, (2)



**Fig. 15.** Persistence factors. Data was given in 110 articles. The total number of values is 210 because some articles provided separate values for multiple categories.



**Fig. 16.** Dropout factor categorisation.  
Source: Redrawn from [122, p. 439].

course factors, (3) environmental factors, (4) instructor factors, and (5) social factors.

Even though participants' motivations are only a single subcategory in the categorisations above, there are some interesting findings regarding these. In [10], it was suggested that completers and dropouts have different motivations. They also grouped the reasons for enrolling into three themes: personal benefits, educational benefits, and career benefits. Zheng et al. [157] had found four categories of motivations for MOOC participants: satisfying current needs, preparing for the future, satisfying curiosity and connecting with people.

### 5.8. Interventions

33 articles (15.3%) described and tested a specific intervention designed to reduce dropout. The strategies employed were diverse, targeting both student behaviours and the learning environment itself, as classified in Table 15.

An analysis of the **types of intervention** reveals two main areas of focus. The most common category of intervention targets *learning provider factors* (22 instances), which focused on modifying the course or platform. These studies frequently explored environmental changes

**Table 13**  
Dropout factor classification.

| Category                        | Subcategories                                                                                                  |
|---------------------------------|----------------------------------------------------------------------------------------------------------------|
| Learning provider factors       |                                                                                                                |
| E-learning environment          | Quality of environment<br>Social factors<br>Support and feedback (from system, instructors, or peers)          |
| Course content and organisation | Quality of content<br>Difficulty<br>Workload<br>Content timing                                                 |
| Other                           | Credit mobility<br>Cost                                                                                        |
| Student factors                 |                                                                                                                |
| Time                            | Time available<br>Amount of effort<br>Timing of effort<br>Support from student's environment                   |
| Motivation and attitude         | Motivation (including relevance)<br>Satisfaction<br>Learning style<br>Collaboration<br>Bad earlier experiences |
| Prior capabilities              | Subject knowledge<br>Computer and e-learning skills<br>Language skills                                         |
| External factors                | Technical issues                                                                                               |
| Other                           | Health<br>Demographic aspects<br>Performance                                                                   |
| No cause found                  | n/a                                                                                                            |

such as adding *support and feedback* mechanisms (5), implementing *gamification* (5), creating *adaptivity* in the learning path (4), improving the *quality of content* (3), having suitable social features (2), or focusing on technical difficulties (1).

Equally prominent were interventions targeting student *motivation and attitude*, appearing in 19 instances. Within this category, simple *encouragement* (for example, personalised messages or reminders) was the most frequently studied strategy (9 articles), followed by interventions that involved *collaboration* (4) and *rewards* (4).

This distribution shows a balanced research interest between psychological “nudges” aimed at the learner and structural changes to the learning experience itself.

Regarding the **effect of the interventions**, they were mostly successful. The effect in the following is reported as the arithmetic difference between two percentages (percentage points); for example, a dropout reduction from 40% to 30% is a 10 percentage point effect. The percentage point values of successful interventions were typically between 3%–30% (12 cases). Only two reported higher values. Of these, in the article by Shen et al. [137], with a reduction from 81% to 24%, or 57 percent points, this was achieved by first effectively guiding students through the initial steps of starting a MOOC and helping them adapt to the MOOC setting. Students received this information through a text-based notice. Second, the progress of the students was monitored and they were encouraged to continue learning if any problems were observed. In the second article with a high level of success by Sureephong et al. [127], non-monetary rewards were used as an intervention. There were three different rewards used: premium coffee mugs as a tangible reward, a letter of commendation from the CEO as a social reward, and training programmes as a job-related reward. In the tangible reward group, dropout was only 37%, while in the worst group, with training programmes as the reward, the dropout level was 84%, a difference of

**Table 14**  
Persistence factor classification.

| Category                        | Subcategories                                                                                                                                                      |
|---------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Learning provider factors       |                                                                                                                                                                    |
| E-learning environment          | Quality of environment<br>Social factors<br>Support and feedback (from system, instructors, peers, or via assessments)                                             |
| Course content and organisation | Quality of content<br>Workload and duration<br>Additional material<br>Gamification<br>Course description<br>Quizzes<br>Localisation<br>Adaptivity<br>Communication |
| Social factors                  | n/a                                                                                                                                                                |
| Other                           | Credit mobility<br>Cost                                                                                                                                            |
| Student factors                 |                                                                                                                                                                    |
| Time                            | Time available<br>Amount of effort<br>Timing of effort                                                                                                             |
| Motivation and attitude         | Motivation (including relevance, reward, certification, and encouragement)<br>Satisfaction<br>Learning style<br>Collaboration                                      |
| Performance                     | n/a                                                                                                                                                                |
| Prior capabilities              | Subject knowledge<br>Computer and e-learning<br>Language skills<br>Earlier grades                                                                                  |
| Other                           | Health<br>Demographic aspects                                                                                                                                      |

47 percentage points. The CEO's note of appreciation was only slightly better than training programmes with a dropout rate of 77%.

When considering other interventions where the effect was 20 or more percentage points, in [146] content and communication improvements reduced dropout by 30 percentage points from 80% to 50%. [166] used “peanut effect” as the most efficient intervention (the term was derived from the fact that people are more willing to gamble when playing for “peanuts”, or a small outcome). The effect had been achieved by lowering the number of questions at the beginning of the course and then gradually increasing the amount. This had a positive effect of 26 percentage points (from 86% to 60%). Pre- and post-tests were used as an intervention in [74] and had an effect on dropout of 21 percentage points (from 55% to 34%).

Regarding these most effective interventions, it is interesting to note that they belong to different categories: communication, encouragement, rewards, and assessments. On a good note, these can be added to a course by the course provider. It is equally interesting to note that there are categories in which the effect on dropout was low or zero, most notably adaptivity (although in one article a reduction of 13.9 percentage points was achieved), gamification, collaboration, and social features.

Only in some cases was no effect noticed [97,118,162,164,167]. To be more precise, in [97], a survey email caused students to revisit the course earlier but did not change the actual persistence rate. Likewise, in [168], using a human face was found to have a positive impact on persistence, but the results were not statistically significant. In [128], the intervention was successful only in the first module of the course, and in [95] a reward caused students to focus more on the sections of



**Table 15**  
Intervention classification.

| Category                        | Subcategories                                                              | Count |
|---------------------------------|----------------------------------------------------------------------------|-------|
| Learning provider factors       |                                                                            |       |
| E-learning environment          | Support and feedback (from system, instructors, peers, or via assessments) | 5     |
|                                 | Social factors                                                             | 2     |
|                                 | Quality of platform                                                        | 1     |
| Course content and organisation | Gamification                                                               | 5     |
|                                 | Adaptivity                                                                 | 4     |
|                                 | Quality of content                                                         | 3     |
|                                 | Communication (e.g. course description)                                    | 2     |
| Student factors                 |                                                                            |       |
| Motivation and attitude         | Encouragement                                                              | 9     |
|                                 | Collaboration                                                              | 4     |
|                                 | Reward                                                                     | 4     |
|                                 | Motivation                                                                 | 1     |
|                                 | Learning style                                                             | 1     |

the course that gave rewards, although there was no difference in the dropout rate.

A negative effect of an intervention was experienced in the use of peer reviews in a course [169].

In one case, the *intervention* itself was by design negative, intentionally worsening the learning experience [170]. In it, the possibility of giving immediate feedback was simply removed.

### 5.9. Findings from qualitative studies

Purely qualitative articles, or articles that had a large qualitative section<sup>7</sup> (altogether 16 articles), concentrated mainly on finding possible causes of dropout or persistence. These studies used interviews (n = 9) or questionnaires with open-ended responses (n = 9) as their data sources. It is worth noting that one of these [171] had interviewed professors, the other articles had students as a focus group. The dropout and persistence reason categories are shown in Tables 16 and 17, respectively.

In the findings, *time-related limitations* were the main reasons for dropping out. This emphasises the challenges for learners to balance course demands with other commitments. The amount of time-related problems also resonates with the findings of this article. However, as a warning, simply asking students why they drop out is problematic, as it is possible that some cite time or workload as a convenient excuse. Such self-reports can be biased by factors such as motivation or content *difficulty* and thus do not reflect the true causes of dropout [79, p. 25]. Dropout was directly associated with low student *motivation* in four cases. In [171], this was more specifically linked to the lack of external pressure. The lack of interactivity mentioned in [142,172] may also be related to motivation. From another point of view, not surprisingly, motivation was found in four cases to be an important factor that supports the opposite of dropout, persistence. *Social factors* also received three mentions as a dropout factor. These manifested as feelings of isolation, lack of integration, or as no sense of community or awareness of others.

Regarding specially interesting findings in qualitative articles, not discussed elsewhere, interviews had shown that selecting only interesting or relevant pieces (“drop in”) can be a factor that causes dropout from the point of view of the course organiser [93]. Still focusing on the student, the main finding of [104] was that it is important for the participant to know what they want from a MOOC themselves.

<sup>7</sup> For example, if a mixed methods study had a large qualitative section, it was included.

**Table 16**  
Dropout reason categories (qualitative).

| Reason category        | Count | Examples      |
|------------------------|-------|---------------|
| Time available         | 8     | [79,91,108]   |
| Motivation             | 4     | [91,157,171]  |
| Difficulty             | 3     | [90,91,124]   |
| Social factors         | 3     | [157,171,172] |
| Language skills        | 2     | [91,173]      |
| Technical issues       | 2     | [91,173]      |
| Learning style         | 2     | [91,94]       |
| Relevance              | 2     | [93,173]      |
| Support and feedback   | 2     | [171,172]     |
| Quality of environment | 2     | [157,172]     |
| Workload               | 2     | [124,157]     |
| Other                  | 4     |               |

**Table 17**  
Persistence reason categories (qualitative).

| Reason category      | Count | Examples      |
|----------------------|-------|---------------|
| Motivation           | 4     | [10,104,142]  |
| Support and feedback | 3     | [108,121,126] |
| Quality of content   | 3     | [108,121,124] |
| Learning style       | 2     | [92,124]      |
| Collaboration        | 2     | [90,174]      |
| Other                | 9     |               |

### 5.10. Pedagogical implications

The pedagogical implications of the articles can be grouped into five thematic areas, as discussed below and as shown in Table 18. As a special note, the persistence causes discussed in previous sections are not repeated here.

**Course content** The *real-world applicability* of course content became an important factor in this theme, both with respect to the needs of adult learners [9] and the general population of MOOCs [141, 175]. In addition, *current and high-quality courses* were mentioned as persistence factors [136,172,176].

*Linguistic accessibility* [119,172] and access to *additional learning resources* [107,124] were also discussed. Finally, other mentioned themes were an average *difficulty*, a moderate *workload*, a *duration* of less than eight weeks [124], and also *adapting the content* to the needs of the students [66].

**Instructional design** As a theme, instructional design had the most pedagogical implications. In here, *student engagement monitoring and intervention* emerged as an important category. Many of these in a quite straightforward manner suggest monitoring student engagement, as it will help to find struggling students [76,

119,147,177–180]. Interventions in a general sense were also suggested [142,166]. In some cases, the recommendations are more exact, for example, by finding students who have only positive comments on the forums (or especially those who have some negative and mostly positive inputs) [181], or those who do not submit their initial activities [182]. An approach is to find what resource access type a student belongs to (that is, how a learner accesses the available learning resources, for example, is the content followed in a scattered fashion or alternatively in the order in which they are presented) and react accordingly so that the discovered learning style is best supported [107]. If an intervention is made ad hoc, it is possible that it has little value [120].

In addition, *social learning design* was an important category. Again, several articles directly suggest to organise a way to have social connections within the learning environment. The solutions include blogs [89], synchronous and asynchronous forums, games, message boards, Dropbox for extra material sharing [172], text or video chats, small-group projects, facilitated discussions [157], and wikis [124]. In a software development context sharing code was mentioned [183]. In one case it was suggested to make connecting and engaging with others mandatory [184]. Also just having a community was seen as valuable in itself [99,185]. As a special case of social learning, collaborative learning was seen as a suggested way to increase student persistence [99]. Creating collaborative learning groups should be done with care, though, for example, grouping active and non-active learners can cause issues [94]. Another special case is peer learning [184].

Regarding *flexible learning design and self-regulated learning (SRL)*, almost all of the excerpts just mentioned that such approaches would be beneficial without further advice [12,68,99,142,157,165,172,175,184]. SRL was seen valuable in one case [82], but warned against in another in that it would not necessarily result in better learning retention [116].

The category *assessment and feedback design including quizzes* included several approaches. Allowing more opportunities to try quizzes and assignments was suggested in two cases [66,186], and a trial-and-error strategy was even encouraged in one [62], although the latter also warned that students may use the automated check tool instead of actually compiling code themselves. The use of quizzes and assessments to diagnose and guide learning was found to be beneficial in two separate articles [63,107]. In two cases, the quizzes were directly linked to videos: students were asked to answer questions immediately after watching each video, and when a quiz question was incorrectly answered, the platform recommended reviewing the relevant video content [107,176]. In-video quizzes were suggested in another case [186]. The simple requirement of having a final assignment was mentioned once [124]. Finally, regarding assignments, some things to avoid were mentioned, namely consisting of simple tasks intended to fill the students' time, testing only rote memory, or having confusing instructions [141].

The articles offered several concrete suggestions regarding the design of *motivational features and gamification*. First, incentives such as badges, awards, and certificates were mentioned [187,188]. Second, gamification was suggested [20,94,188], although the value was unclear in one case of professional learning [188]. As single mentions, ideas of comparing students between each other [157], explaining how students of previous course instances had learnt [66], tuning automatic grading to give more weight to forum posts in the middle and late phases of a course [189], and generally simply to encourage students to stay even if they encounter difficulties [62]. Lastly, various

motivational activities and resources were highlighted, such as reflective journals and questions, practical exercises, problem solving, simulations, peer assessments, short videos, and real-life examples and assessments [99].

The basic idea behind *content-path customisability* is to allow learners to build their own learning from the most relevant parts of a course or even several courses [13,119,172]. This can also be related to customising the learning level [100], starting at different times or locations in a course [119], or even providing suggestions for resources from upcoming chapters [107].

Several ways to implement *multimodal content delivery* were offered. Concrete examples include enriched learning components such as mind maps, interactive animations, integrated development environments and simulators [89], complementing videos with multimedia such as hyperlinks, text, image, diagrams, and animations [172], diversifying video formats and seamlessly integrating each video with follow-up questions [176], context-orientated discussions, reflective journals and questions, practical exercises, problem solving, simulations, peer assessments, short videos, and real-life examples and assessments [99] (last of these also related to motivational design above). Furthermore, an article suggested providing the option to complement or even replace videos with text or diagrams and allowing users to choose the content media [172]. A fairly specific hint about the length of the videos instructed the videos to be short, even shorter than six minutes [141].

*Early stage support* can be provided in multiple ways: feedback, encouragement, early dropout identification, and related support [182]. The timing should be early in the course [182], even specifically during the first three weeks [189]. On the topic of the use of learning management systems (LMS), it is recommended to provide students with support materials, such as instructional videos or orientation sessions, to help develop the necessary technological skills [99,128]. In addition, offering information on common technical difficulties and guidance on how to overcome them is useful [128].

*Structured content progression* offers several views to support learning. First, pieces of the course, such as videos and related questions, should be tightly combined [176]. Second, special attention should be paid to bridging learning between course weeks [103] and topic transitions [64]. The latter can be achieved, for example, by implementing cliffhangers. Third, it was recommended to pay special attention to completing assessments at the beginning of the course [182].

Comments about *active learning design* consisted of two cases of mentions about including a project in the course [10,124]. More precise project-related information included the need for help starting a project, possibly making the project voluntary, and adding new learning to the project [68]. Promoting active learning can be achieved, for example, by embedding collaborative tasks, peer-led presentations and assessments, interactive discussions, and timely instructor feedback so students engage directly with each other, with the teacher, and with varied course content [99]. However, it was also observed that the introduction of active learning strategies can generate additional problems for learners [94].

Regarding *cognitive load and attention*, the load related to learning both the e-learning environment [99] and the course topics simultaneously warrants attention, possibly by separately training participants about the use of the platform [128]. In a similar sense, it should be avoided to promote too many learning resources at once [107]. As separate comments, prepopulating forums [141], and creating fewer more challenging problems instead of a series of easy problems were reported [150].

There was also an excerpt on quizzes and exams; designing them as a “minimum-cost attention network”, so that learners at risk are not cognitively overloaded and learning remains cost-effective [190].

*The degree of freedom of scheduling with a fixed schedule* was considered beneficial as it allows collaboration between learners [12]. A fixed schedule was also suggested to suit learners who are preparing for some future job or other education [157]. In other cases, it was mentioned to help retain learners [191], or was just generally recommended [63,124].

On the other hand, *the degree of scheduling freedom with an open schedule* was considered suitable for learners who cannot work with the same schedule as others [12,157]. Or an open schedule might be beneficial if learners simply do not want to wait to access materials they want to see [149].

*Learner onboarding and expectation setting* is an interesting category, related to early-stage support but separate from it. As concrete examples, a visual diagram of the course content that includes the expected time and effort [172], a syllabus showing the learning path [63], or a welcome tutorial on guiding novices in MOOCs, their function, and structure [136]. The registration pages could also highlight the problems faced by students who fall behind [103]. Or there could be a section that forces students to become aware of their internal motivation to complete a course [175].

There are many factors that help to design a course that is *inclusive and culturally responsive*. Examples include vital components of culture, such as language, symbols, norms, beliefs and values [119] or background, value systems, current situations or personal objectives [157]. Ways of taking these into account were suggested to include the formation of subgroups [157] or project groups [10], or simply using a map of the distribution of learners around the world [157]. In addition, community building through the creation of a student list and their performance or even the display of the number of students online at any given time was suggested [157].

*The instructor-led scaffold* included ideas for recommending learning paths based on similar successful learners [107], supporting students without external encouragement or having personal or work difficulties [76], proactively anticipating problems and acting accordingly [94], and providing constructive suggestions for problems that arise in interactions with students [100].

Three articles suggested using *gradual learning progression* to retain students. Especially, the first investments should be as small as possible, for example, via smaller assignments [191]. On a more general level, gradually increasing the difficulty of exams was recommended [164]. As a suggestion related especially to computing education, a gradual learning curve would assist in adapting to computational thinking before having to learn a programming language [62].

Regarding *the support for delayed students*, simply creating a separate discussion area was a suggestion to help them [103]. Another, although requiring more effort on the part of the course providers, was to offer simplified material as support [59]. In addition, arranging peer review tasks so that latecomers can also join those was suggested [169].

**E-learning environment** The e-learning environment provides services that can be used to create course content and instructional design. Most of the recommendations in this area focus on strengthening the social presence and the supporting communication infrastructure [89,90,157,175]. Of these, *student–student interaction design* has some concrete examples in providing a

reputation or a badge system [141,183], interaction between current and former learners [188], utilising gamification [188], or improving emotional communication by providing occasional synchronous classes [192]. On the topic of *communication timing and rhythm*, there is a suggestion of promoting communication also during weekends [147], and also by providing certain timed meetings or classes to learn more about possible problems the participants have encountered [100].

Several notes have been collected on the *design and facilitation of discussion forums*. Again, some concrete ideas include opening a forum already before a course starts [90], using content-orientated discussions [99], and organising forums by language, country, or course topics [136]. In addition, the importance of instructors and teaching assistants to participate in forums was highlighted. The format of discussions can also be as questions and answers [176]. Finally, it is suggested that instructors use tools to understand the dynamics of forum discussions [189].

*Reminders* as a platform service can include notifications of new content, schedules, or deadlines [149,191,193]. Generally, those have been found to be effective; however, in one case simply sending a weekly email reminder about tasks and new forum content was not seen as useful [147]. A completely different platform service, *intelligent tutoring* – possibly powered by artificial intelligence – was also suggested and considered an important factor for persistence [60,88].

The last aspects of this category concern *accessibility*. First, the course platform should ensure that it can operate under varying connectivity conditions. Second, videos are suggested to have captions or transcripts. Complementing these with diagrams and other similar content from videos would ensure that various devices used by students are better served (and also assist with the aforementioned connectivity conditions) [172].

**Instructor** As a theme, “Instructor” did not get so many excerpts in any of its categories that it would be included in Table 18. Regarding related findings, interestingly, the *affective presence* was quoted as an important factor [141,192], along with *being aware of any misconceptions* [65]. Regarding the affect, there was a warning that not smiling or looking out of the camera should be avoided [141].

Otherwise, the mere *presence of an instructor* was mentioned as influential in persistence [141]. However, simply reading the slides aloud or presenting irrelevant information was found to be a hindrance to satisfaction [141].

**Administrative factors** Regarding course management, *being able to preview the content of the course* was mentioned in the excerpts as a factor in reducing the number of people who register for a course just to see what it contains [13,103,156]. In addition, the possibility of deregistering when losing interest was suggested [60]. *Credit recognition* was seen as influential, especially for employee learners [128,188], but also for students [124, 157]. Finally, it was mentioned that course organisers should try to *mobilise dropouts for upcoming courses* [191] and also highlight the possibility of coming courses for learners who register late [59].

In learner modelling and adaptive response, the first suggested step would be to try to *recognise the intents of students* [13, 165,187]. This is also close, but somewhat different from the suggestion of *affective monitoring* of students (including confusion) [57,194]. Continuing on the same lines as monitoring is *the detection and prediction of dropout* [177,195,196]. And when it occurs, it would be advisable to *diagnose any factors related to dropout* [76,147].

**Table 18**  
Summary of key pedagogical and design implications.

| Theme and related categories                                            | Count | Examples           |
|-------------------------------------------------------------------------|-------|--------------------|
| Course content                                                          |       |                    |
| Quality of content                                                      | 3     | [136,172,176]      |
| Real-world application                                                  | 3     | [9,141,175]        |
| Instructional design                                                    |       |                    |
| Student engagement monitoring and intervention                          | 13    | [120,166,181,197]  |
| Social learning design (incl. collaborative learning and peer teaching) | 12    | [89,94,99,172,183] |
| Flexible learning design and self-regulated learning <sup>a</sup>       | 11    | [68,82,172]        |
| Assessment strategy and feedback design (incl. quizzes)                 | 9     | [62,107,150,186]   |
| Motivational feature design (incl. gamification)                        | 9     | [94,157,189]       |
| Content-path customisability                                            | 7     | [13,107,172]       |
| Multimodal content delivery                                             | 7     | [89,99,172]        |
| Early-stage support                                                     | 6     | [107,128,182]      |
| Structured content progression                                          | 6     | [64,176,182]       |
| Active learning design <sup>b</sup>                                     | 5     | [10,99,175]        |
| Cognitive load and attention related design                             | 5     | [107,128,190]      |
| Degree of scheduling freedom: fixed schedule                            | 5     | [124,157,191]      |
| Degree of scheduling freedom: open schedule                             | 5     | [157,175,191]      |
| Learner onboarding and expectation setting                              | 5     | [63,172,175]       |
| Inclusive and culturally responsive design                              | 4     | [10,119,157]       |
| Instructor-led scaffolding                                              | 4     | [94,100,107]       |
| Gradual learning progression                                            | 3     | [62,120,191]       |
| Support for delayed students                                            | 3     | [59,103,169]       |
| E-learning environment                                                  |       |                    |
| Student–student interaction design <sup>c</sup>                         | 7     | [183,188,192]      |
| Forum discussions                                                       | 4     | [90,136,186]       |
| Reminders <sup>a</sup>                                                  | 4     | [147,149,191]      |
| Administrative factors                                                  |       |                    |
| Credit mobility and credit recognition                                  | 4     | [124,128,188]      |
| Enrolment flexibility and easy access                                   | 4     | [59,103,156]       |
| Dropout detection and prediction                                        | 3     | [177,195,196]      |
| Intent recognition                                                      | 3     | [13,165,187]       |

<sup>a</sup> One did not have an effect.

<sup>b</sup> One negative effect.

<sup>c</sup> Also with former students.

In summary, the pedagogical implications observed and presented above can be broadly grouped according to the primary stakeholder addressed. Several recommendations are aimed at instructors (for example, engagement monitoring and early stage support), while others are targeted to course designers (such as content path customisability, assessment strategy, inclusive course design, and multimodal content delivery). Additional implications, including platform accessibility and technical support features, are primarily relevant for platform developers.

### 5.11. Suggested further research in articles

We categorised 142 suggestions for further research given in the set of articles published in the past five years. Applying the TCCM future research reporting framework [198], most of the statements were related to “Characteristics” (61%), then came “Methodology” (24%) and “Context” (14%), and “Theory development” had only two suggestions (1%). It was interesting to see that more than half of all categorisations were related to the suggested “Characteristics” – in other words, what should be studied, not how. The most important categories in these larger themes are shown in Table 19. Regarding the suggestion count mentioned above, it is worth mentioning that ten of the suggestions explicitly listed future steps that the article authors themselves want to take, not general calls for action.

Now, it is tempting to concentrate only on the categories with the highest number of excerpts. However, when reading the actual texts we feel that such an approach would result in a quantitative fallacy [199,

p. 90], in other words we would report as important something that was easy to quantify and count. So, apart from analysing the factors with highest counts, we have extracted the most interesting suggestions, and also will highlight what may be missing from the results.

**Characteristics** Most of the suggestions are related to *motivation*.

There are quite a few comments on motivation sources, such as external factors such as the job market, employment possibilities, wages, family incentives [200] and social drives [77], and internal factors such as personal goals [68,69,77,100]. Continuing on the same topic, one article added the view of how the difficulty of the topic affects motivation [150]. The manifestation of motivation to engage was also mentioned [192]. An article also added the motivation of instructors to the list [19]. Still related to motivation, various incentives have been suggested [25, 76,188,201].

In the category *continuation with additional inputs* (meaning some completely new types of input), a combination of trace and self-referential data was suggested [164,180]. It is easy to see that this approach strengthens the value of both data sources.

The difficulty of *course content* is a feature often overlooked, but it is understandably having an effect on dropout [150,166]. In [166], the duration of a course was also mentioned as a feature.

The effect of *instructors* was highlighted in two cases, again an understandable feature that affects dropout [19,100].

Ways of clustering the students in regard to *learning* as how completers and non-completers differ were highlighted [66], as well as why certain implementation of *support* works for some students but not for others [75]. Another important aspect of *learning* is the cognitive load imposed on the students [150]. And a broad understanding of learners is also beneficial in *forum participation* – topic-unrelated posts may provide additional information for course design and facilitation, for example regarding complaints or technical difficulties [189].

Our final actual comments are about *new suggested outcomes*, however, the actual important point is the call for validated measures of learner performance [189]. We would of course like to extend the same view to all aspects surrounding dropout phenomenon.

Lastly, in ten cases there were no concrete ideas for the characteristics, only a comment that more of those would be needed.

**Methodology** In the category *change of used method*, the need for more qualitative approaches was highlighted in several sources [25, 49,88,175,202]. In [175], interviews were also specifically mentioned.

**Context** No specifically noteworthy suggestions.

**Theory** Regarding *theory development or extension*, we want to highlight the lack of calls for future research in this category. The low amount of theory-related comments really jumps out of the data – we would like to interpret this as an urgent need for more theory in the field.

## 6. Discussion

### 6.1. Summary of findings

This systematic mapping study synthesised 216 articles published between 2007 and 2023 to map the research landscape on dropout in tertiary-level continuing professional development (CPD) e-learning



**Table 19**  
Suggested further research in articles.

| TCCM category      | <i>n</i> | %  | Top categories                                                                                        | Examples                                               |
|--------------------|----------|----|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Characteristics    | 86       | 61 | Motivation (16)<br>Student factors (10)<br>Instructional design (8)<br>(Unknown factors (10))         | [143,150,192,200]<br>[75,77]<br>[162,203]<br>[145,204] |
| Methodology        | 34       | 24 | Prediction enhancements (9)<br>Change of used method (8)<br>Model refinement (8)<br>Time dynamics (7) | [205,206]<br>[25,202]<br>[189,207]<br>[49,99]          |
| Context            | 20       | 14 | Generalisation and broader testing (17)                                                               | [118,123,143,208]                                      |
| Theory development | 2        | 1  | Theory development or extension (2)                                                                   | [162,192]                                              |

courses. The findings provide detailed answers to our guiding research questions.

The research question 1 was as follows: “What has been researched regarding dropout in tertiary-level continuing professional development e-learning courses?”. Of the whole set of 216 articles, 81 had a research question about how some factors, decided beforehand, have an effect on dropout. 74 articles focused on prediction using machine learning techniques. The next categories that dealt with some kind of remedies against dropout (remedies, interventions, and experiments) were already quite rare, with about 20–30 articles each. It was somewhat surprising to see that research questions trying to find out the causes of dropout had a count of only 20. Considerably less attention has been paid to explicitly developing theoretical models (8 articles). The qualitative studies identified ( $n = 16$ ) sought to uncover causes of dropout or persistence through methods such as interviews and open-ended questionnaires, sometimes revealing nuanced learner behaviours such as selective engagement (“drop in”), which challenges traditional definitions. Furthermore, while our scope was tertiary-level CPD, only a small fraction of studies (10 articles) explicitly centred their research questions on adult learners in this specific context. Bibliographically, following low activity prior to 2013, research output increased markedly, peaking in 2019 (32 articles), likely driven by interest in MOOCs. Since then, activity has settled at a slightly lower but still elevated level, averaging over 20 articles per year.

The research question 2 asked “How has the research been carried out? What have been the research contexts, target groups, and research methods?”. The type of courses included mainly MOOC courses with 90%<sup>8</sup> of the articles, online courses represented 10% of the set. The large amount of MOOCs is understandable due to a large amount of existing datasets recording MOOC participation. Such data sets were used for a large part, or 43% of the MOOC-related articles (contrasting with 14% on the online course articles). Interviews and questionnaires were used in 24% of all articles and a similar prevalence was observed for courses that researchers can directly access or even modify for research purposes (26%). Methodologically, the field relies heavily on quantitative research designs (88%), analysing readily available platform data such as clickstreams, recorded activities, and course results, although questionnaires and demographics were also common. Qualitative methods such as interviews (only ten studies) remain scarce. There is also a significant lack of theoretical grounding, with theories, models, frameworks, or methods explicitly used (whether applied, extended, or newly created) in only about 19% of non-prediction-focused articles.<sup>9</sup> Although MOOC participants were typically adults

with high educational levels, detailed reporting on course context (workload, prerequisites, target groups) and participant demographics was often inconsistent or missing. Common course subjects included ICT, education, STEM, business, and arts/humanities, although many studies lacked clear subject information.

The research question 3, “What results related to dropout have been found?” was split into two sub-questions as follows.

The research question 3.1 was: “How has dropout been defined and how should dropout percentages be interpreted from this perspective?”. The review confirmed that defining and measuring dropout is complex. Operational definitions varied, particularly with respect to the baseline used for calculation (e.g., total enrolled vs. those who started activity) and the completion criteria (e.g., activity thresholds or grades). High MOOC dropout rates ( $Mdn = 85.5\%$ ) appear inflated when calculated from enrolment numbers, largely due to “no-shows” and “samplers”, the latter of whom only want to see what the contents are like. However, even when calculated from active starters, the rates in this review remained high ( $Mdn = 82.0\%$ ), contrasting with the lower rates observed in other online courses ( $Mdn = 55.0\%$ ). This suggests that high attrition is intrinsic to the format and not solely an artefact of measurement. Furthermore, the traditional concept of “dropout” may not fully apply to CPD learners who might achieve personal learning goals without formally completing a course (representing a “voluntary departure” rather than failure). Various learner typologies such as samplers, completers, and casual learners reflect these diverse intentions.

The research question 3.2 was: “What have been found as the reasons for dropout and what kind of interventions have been carried out to diminish it?”. Our review has the following findings.

- Factors influencing dropout and persistence: Quantitative analysis revealed time constraints as the most frequently cited factor contributing to dropout (34% of mentions), followed by issues related to motivation and attitude (18%). In contrast, for persistence, motivation and attitude was the leading factor (26%), closely followed by available time (22%). This highlights an interplay: time is the primary barrier, while motivation is the primary driver, and having sufficient time is a key enabler for persistence.
- Qualitative insights on factors: Qualitative studies corroborated time limitations as the main stated reason for dropout, but cautioned that this might sometimes mask underlying issues like motivation or content difficulty. These studies also reinforced the importance of motivation (both its lack for dropout and presence for persistence) and highlighted difficulty, or social factors such as isolation as contributors to dropout.
- Interventions: Research on interventions was relatively sparse (33 articles). The interventions targeted both student motivation (e.g., encouragement, rewards, collaboration) and course/platform design (e.g., feedback, gamification, adaptivity). Most showed some positive effect, but success varied widely; some interventions had limited, inconsistent, or even negative impacts. Simple, well-timed strategies like guidance and personalised encouragement, or tangible rewards sometimes yielded significant dropout reductions (up to 57 percentage points), while more complex approaches like adaptivity or gamification often showed limited or inconsistent effect.

A high-level summary of these factors, along with their corresponding pedagogical implications, is presented in Table 12.

## 6.2. Contribution in relation to previous reviews

Although the present systematic mapping study shares the broad topic of e-learning dropout with the 12 review articles identified in our search (Table 1), its specific focus on tertiary-level CPD e-learning and its mapping methodology produced a more detailed picture of the research findings within this particular domain. Most existing reviews

<sup>8</sup> We have used zero decimals for percentages in the discussion section.

<sup>9</sup> We have omitted articles that have only machine learning prediction or other prediction as the research question classification (66 articles), meaning that the percentage is calculated from 150 articles.



concentrate broadly on MOOCs, specific technical aspects such as machine learning prediction (where the focus was often on identifying factors relevant to those predictive models [15–17]), or particular influencing factors such as engagement, motivation, or gamification [18–20, 25]. We compare the results on the basis of key thematic areas below.

On the theme of **dropout definitions and rates**, the findings in previous reviews frequently highlighted the notoriously high dropout rates in MOOCs. Several reviews (five out of the twelve) cited specific completion figures, typically below 15% [17,20,22–24]. Some of the previous reviews acknowledged the challenge posed by high initial enrolment versus active participation in MOOCs, as follows. Four out of the twelve reviews illustrated this with concepts such as learner typologies or the “funnel of participation” (visually depicted as part of the AMOES model in Fig. 13) [14,20,23,25], and one noting that the resulting “noisy data” requires focusing on active learners [17]. However, in these reviews, a deeper analysis of the complexities of defining dropout versus selective learning was very limited. In contrast, the findings of this mapping study provide a more detailed synthesis of how operational definitions impact reported rates (Section 5.6). We confirmed the high dropout in MOOCs when calculated from initial enrolment (often >80%), but also highlighted that rates calculated from active starters were only marginally lower, remaining distinct from the non-MOOC online courses in our set (medians 37%–55%). Building upon the awareness of definitional complexities present in some prior reviews as discussed above, our mapping study provides a detailed synthesis of the discussions found in the literature regarding diverse learner intentions and alternative interpretations of non-completion. We collected findings on concepts such as “samplers”, “casual learners”, “voluntary departure”, and “closure”, perspectives which are particularly relevant for interpreting dropout rates among the adult learner populations prevalent in MOOCs and CPD contexts, for whom achieving specific goals could constitute success irrespective of formal course completion.

Regarding **dropout and persistence factors**, the factors identified in the previous 12 reviews broadly aligned with the findings of our own quantitative analysis. Regarding *dropout*, in both datasets, time constraints and motivation/attitude consistently emerged as the most frequently cited factors associated with dropout. Our analysis showed that both time and motivation had 10 mentions as sources for dropout (among the 8 providing dropout factor data<sup>10</sup>), followed by prior capabilities (7 mentions) and course content/organisation (6 mentions). Similarly, our full analysis found time (34% of mentions across 69 articles addressing dropout factors) and motivation (18%) as the top two dropout factors. For *persistence factors*, motivation/attitude was the leading category in both analyses (accumulating 10 mentions in the 8 reviews providing persistence data, and accounting for 26% of mentions across 110 articles addressing persistence factors in our study). Aspects related to course content/organisation were also prominent in both sets (8 reviews, 15% in our study). Indeed, specific reviews within our target article set delved particularly deeply into motivation or engagement as key factors [18,19,25].

However, a significant difference emerged regarding the role of time as an enabler of persistence. Although it was the second most cited persistence factor in our quantitative analysis (22% of the mentions), it was notably infrequent in the analysed subset of reviews (only 2 mentions). Interestingly, this quantitative finding was further nuanced by our synthesis of the qualitative studies (Section 5.9). Although lack of time was the most frequently cited dropout reason in the qualitative data (8 cases of 23 total), having sufficient time did not emerge as

a persistence factor in those studies at all. This suggests that while time constraints are clearly highlighted dropout factors, qualitative explorations of persistence may focus more on active motivational drivers, or positive course experiences rather than the mere availability of time.

Lastly, regarding **interventions**, in previous reviews interventions were not the main focus of most of the 12 reviews. Where discussed, they were related to specific techniques such as gamification [20] or related to the application of data science and prediction to trigger support [14]. In general, a systematic overview of the types and effectiveness of interventions across the board is lacking. On the other hand, our mapping study catalogued the types of interventions (e.g., encouragement, feedback, adaptivity, rewards, gamification) tested in 33 primary studies and summarised their reported effects (Section 5.8). This provides a structured overview specific to the CPD context, revealing the variety of approaches attempted and their inconsistent outcomes. In particular, our analysis highlighted that simpler, targeted interventions, such as personalised encouragement or specific rewards, sometimes yielded substantial dropout reductions, while more complex technological solutions such as adaptivity or gamification showed more mixed results in the reviewed studies.

### 6.3. Research gaps

Several **research gaps** and underexplored areas were identified in the reviewed literature, as outlined below.

#### • Research methods

- Qualitative and theoretical approaches remain rare, with most studies employing quantitative methods
- Theory building is uncommon, both in terms of developing new models and applying existing frameworks

#### • Research focus and coverage

- Few studies address non-MOOC adult online courses, or commercial adult online courses, the latter being almost completely absent
- Examinations of interventions are limited and differ considerably in their design
- Only a small number of articles analysed both dropout and persistence factors, with only 20 studies reporting both types
- Short courses are under-represented, with only 12 articles examining courses of less than five weeks in duration

#### • Reporting

- Reporting of course context, prerequisites, and participant demographics is often sparse, which may hide relevant patterns and complicate the comparison of findings between studies

This review confirmed a clear gap: the nearly total absence of research on dropout within the significant commercial e-learning sector. To build necessary knowledge in this area, policymakers and funding bodies have a potential role, for example, by encouraging or supporting research collaborations and data sharing with commercial providers

### 6.4. Implications for pedagogy and policy

The findings of this systematic mapping review offer several implications for those involved in designing, delivering, and overseeing tertiary-level CPD e-learning, including educators, instructional designers, institutions, and policymakers. These implications arise from both the broader patterns identified in our analysis and from specific suggestions within the reviewed literature.

<sup>10</sup> The mention count for broad categories like “time” or “motivation” can exceed the number of contributing articles because these categories group several specific factors (e.g., “time” includes “time available” and “timing of effort”). An article discussing multiple specific factors within the same broad category was counted for each mention relevant to that category.

First, our systematic mapping study of the research landscape highlights several insights for practice and policy. Suggested by the **dropout and persistence factors** identified in the literature (Section 5.7), a key point is to recognise the importance of time and motivation: The consistent finding that time constraints are the main barrier to completion, while motivation is the main driver of persistence in this context, underscores the need for pedagogical strategies and institutional policies that actively address these core issues. Course designs would benefit from incorporating flexibility and realistic workload expectations, while targeted efforts should focus on boosting and sustaining learner motivation throughout the course, utilising a high-quality and usable e-learning environment. Another important consideration is to understand diverse learner goals in CPD: This review confirms that traditional completion metrics often fail to capture success for CPD learners, many of whom engage selectively to meet specific professional needs (“voluntary departure”) rather than aiming for full course completion. Institutions and course providers should therefore consider offering more modular structures, exploring alternative credentialing (e.g., micro-credentials), and clearly communicating outcomes to better align courses with diverse learner goals. Allowing content previews might also help this alignment.

The varied success of **interventions** (Section 5.8) reported in the literature offers a nuanced lesson for practitioners. Although intuitive, high-effort strategies such as complex gamification or adaptive learning paths showed inconsistent results, often failing to significantly reduce dropout. In contrast, simpler, well-timed “nudges”, such as personalised encouragement, clear guidance, or tangible rewards, frequently yielded substantial positive effects, in one case reducing dropout by up to 57 percentage points. Regarding this difference, it is plausible that complex interventions may accidentally introduce additional cognitive load or technical friction, whereas simple interventions succeed by directly addressing motivational issues or confusion at the right moments without requiring new learning behaviours from the student [94,128,190]. Therefore, a one-size-fits-all approach is unlikely to be optimal. Data analytics should consequently be used not just for prediction, but for diagnosing specific learner difficulties to inform these targeted, low-friction support mechanisms, particularly in the early stages of a course.

Second, in addition to the above implications, the reviewed articles themselves offered numerous **specific suggestions for practice** (detailed in Section 5.10). We present these recommendations following a structure that moves from course-specific elements outwards: starting with course content, then addressing instructional design principles common across courses, followed by the e-learning environment, and finally administrative factors.

Insights regarding *course content* were relatively sparse: the authors emphasised providing high-quality, relevant content with clear real-world applicability.

However, there were quite a few recommendations regarding *instructional design*. Frequent ones included implementing systems to monitor engagement and intervene proactively; fostering social learning through forums, group work, and other tools; offering flexible designs that support self-regulation; and employing effective assessment and feedback strategies, such as frequent, low-stakes quizzes and opportunities for retries. Designing motivational features, allowing content-path customisability and organising multimodal content delivery were also commonly advised. As a special note, some articles recommended a structured content progression, while others emphasised the benefits of customisable learning paths. Similar contrasts were observed with respect to scheduling, with both fixed and open schedules identified as potentially beneficial depending on the context of the course.

On the topic of an *e-learning environment*, strengthening the environment’s support for student–student interaction (e.g., through synchronous classes, badges, and even with former students) and ensuring platform accessibility were highlighted. And regarding *administrative*

*support*, suggestions included providing flexible onboarding and easy access to content, and exploring options for credit recognition where appropriate.

By combining the specific pedagogical suggestions reported in the literature with the overarching insights revealed by this mapping study, such as the key roles of time and motivation, and the need to recognise diverse learner goals, practitioners and policymakers can develop more targeted and effective strategies to support learner success in tertiary-level CPD e-learning.

### 6.5. Limitations of this review

This review has methodological limitations that should be taken into account when interpreting the findings. First, as is typical in mapping studies, no formal assessment of article quality was performed, so the synthesis reflects reported patterns across the literature rather than being weighted by methodological rigour. Second, the categorisation of articles, research questions and factors inevitably involved a degree of subjective interpretation, although unclear cases were discussed between authors to reach consensus. Third, the search strategy focused on the term “dropout”, which may have excluded studies that addressed related concepts such as retention or persistence using different terminology. However, this focused strategy still retrieved a substantial volume of 216 articles, indicating that a good coverage of the phenomenon was achieved. Fourth, the frequencies reported throughout this review largely reflect the number of articles addressing a certain topic or finding, not necessarily the weighted importance or impact of those findings. These counts may also reflect what has been easiest to study (for example, quantitative analysis of readily available MOOC platform data) rather than what is most pedagogically significant. To counter this, our separate analysis of qualitative findings (Section 5.9) aimed to provide deeper insights beyond mere frequency. Finally, it should also be noted that the analysis was limited to factors identified as significant dropout or persistence factors within each article; topics or variables that were studied but were not found to be important, such as instructor-related aspects, were not included in the synthesis.

### 6.6. Bias identification and generalisability

Beyond the methodological limitations of the review process, the characteristics of the included literature introduce biases that affect the generalisability of the findings.

First, there is a distinct **predominance of MOOCs**, which account for the vast majority of the included studies. MOOCs represent a specific pedagogical format characterised by massive enrolments, low barriers to entry, and often low stakes for non-completion. While this review differentiates between MOOC and non-MOOC dropout rates, the vast majority of the qualitative evidence regarding dropout *factors* and *interventions* stems from MOOC environments. Consequently, the findings regarding the causes of dropout are heavily influenced by the specific MOOC context. These insights should be applied with caution to other forms of CPD, such as small-group corporate training or mandatory professional licencing courses, where social pressure and external motivators are significantly stronger.

Second, the literature exhibits a **methodological bias towards quantitative approaches**, particularly those employing educational data mining on log data. While these studies effectively identify *when* dropout happens and *which* interaction patterns predict it, they often lack the qualitative depth required to explain *why* it happens. Furthermore, despite the focus on quantitative metrics, reporting standards vary significantly; as noted in our results, basic data such as participant counts were in many cases omitted, limiting the potential for weighted meta-analyses. This predominance of purely algorithmic analysis may overshadow subtle, human-centric reasons for dropout (for example, changing life circumstances, lack of perceived value) that are better captured in qualitative interviews.

Finally, the **geographical and linguistic bias** of the included literature limits the global applicability of the findings. As indicated by the content provider regions (see Table 11), the literature is dominated by studies originating from Europe and North America. Although East Asian platforms were represented, regions such as Latin America, Africa, and South Asia appeared less frequently. Furthermore, our restriction to publications in English excludes a body of research published in other major languages. Bibliometric studies estimate that while English dominates indexed databases, significant volumes of educational research are published in Chinese, Spanish, Portuguese, and French, particularly in regional repositories such as SciELO (Latin America) or CNKI (China) that are often under-indexed in global databases [209]. Cultural attitudes towards education, persistence, and online learning vary significantly; thus, the dropout factors identified here may not fully reflect the experiences of learners in these non-Western or non-English speaking contexts [210,211].

### 6.7. Recommendations for future research

This systematic mapping review not only summarises the current state of research, but also illuminates several areas requiring further investigation. These recommendations arise from both the significant gaps revealed by our mapping analysis and from explicit suggestions within the reviewed articles. Addressing these will be beneficial for developing a more comprehensive and actionable understanding of dropout in tertiary-level CPD e-learning.

First, based on the suggestions of the literature and the significant **gaps identified in our mapping study** (see Section 6), we recommend the following key directions.

There is a clear need for studies that *extend beyond MOOCs* to include a wider range of online courses. Future research should investigate dropout in:

- University-based CPD courses: While general online university education is widely researched, investigation of dropout dynamics within university-provided CPD courses is rare. These typically target working professionals with different motivations and constraints than traditional degree students.
- The commercial e-learning sector: This almost entirely unresearched area requires dedicated studies, potentially through industry partnerships, to understand dropout in certificate programmes and corporate training
- Short courses: A specific focus on shorter courses would be beneficial, given their common usage in CPD

We also could see a gap regarding *methodological diversity*: the dominance of quantitative methods needs to be balanced. This would include prioritising qualitative studies (e.g., in-depth interviews, case studies, ethnographic approaches) to explore complex experiences, decision-making processes, and contextual factors influencing adult learners. Mixed-method designs could be used to triangulate platform data with learner perspectives. And, importantly, there is a need to address the lack of theoretical approaches, for example, by developing CPD-specific dropout models or by applying and testing established educational theories.

The articles showed an *unsystematic approach to interventions*, which gives limited generalisable results. Therefore, future work should employ more structured designs (experimental or quasi-experimental), with clear descriptions of intervention components and outcome measures. As special cases, (1) it would be interesting to see why simple interventions (such as encouragement) sometimes appear effective while complex ones (such as gamification or adaptivity) show mixed results in the CPD context, and (2) it would be beneficial to develop and test interventions specifically targeting time management and self-regulation support for adult learners with busy schedules.

As there was a prevalence in the article set to analyse dropout factors as correlations, we could see the need to focus on uncovering *causal*

*mechanisms underlying dropout*. Also, the dynamics of persistence should be examined more deeply, understanding the factors and strategies that actively *sustain engagement* over time, rather than only identifying factors correlated with non-dropout. Also, since most of the existing analysis focused on learners, it would be useful to add *other stakeholder perspectives*, for example, those of instructors and course providers.

Our analysis highlighted *inconsistent reporting* across the reviewed articles, which hinders comparability and future synthesis. We recommend that future research adopts more consistent and detailed reporting standards for course contexts (for example, design, workload, subject, duration), platform features, participant demographics (including professional background and motivation) and methodological details.

Second, on the topic of **directions for future research highlighted in the reviewed literature** (see Section 5.11), several themes emerged frequently. Authors often called for a deeper exploration of dropout characteristics, particularly learner motivation (e.g., its sources and incentives). Combining platform data with self-reported data, examining course content and instructor impact, and enhancing prediction models were also common suggestions. Numerous authors also called for more qualitative approaches and testing findings in different contexts. Other potentially important areas that have received less attention were, most notably, explicit calls for new theory development. Such papers were extremely scarce (only two instances in our data pool). This highlights a potential blind spot: the articles frequently suggest incremental work on existing dominant themes (like motivation or prediction) while potentially overlooking less frequent but arguably more fundamental needs, such as building stronger theoretical foundations.

Third, we discuss **opportunities provided by recent developments in artificial intelligence (AI)**. *Findings synthesis* with the assistance of AI has the potential to improve efficiency and even mitigate errors created by a manual analysis process [212–214]. For example, this can mean using AI as an independent second reviewer to verify manual data extraction [215]. Current tools also allow a larger part of the review process to be performed with AI assistance [214]. Practical and formal guidelines for this workflow have been proposed in [213,216]. However, caution is required and results must be verified carefully at each step [217–219]. Tool selection is also important, as different tools have differing profiles in their strengths and suitability [214].

Further potential is provided by AI-assisted *theory discovery*. The opportunities vary from human-in-the-loop Grounded Theory assistance [220] to complex cases where AI tools can generate hypotheses based on patterns and relationships found across multiple articles, which may be hidden or beyond the capability of a human reader [221]. AI approaches can also assist in uncovering causal structures that are traditionally difficult for humans to detect [222]. In fact, there already exist successful implementations of data-driven hypothesis and theory discovery workflows [223,224].

As a separate technique, traditional *eXplainable AI* (XAI) has been successfully used in learning analytics to interpret dropout predictions [225–227], although it was not observed within our primary article set, as these works were published after the time range of our data pool. However, a more recent development is the emergence of *explainable generative AI* (GenXAI) frameworks that combine explanation with automated discovery [228]. An example of this is the *HypoGeniC* framework, which establishes an *iterative hypothesis testing loop* [224]. Applied to dropout research, this framework could be adapted so that a generative model creates an initial dropout theory from a sample of qualitative data, which is then tested against a large-scale quantitative dataset. The system would then find cases where the model made incorrect predictions and route these errors back to the AI, forcing it to recursively adapt and refine the theory until it aligns with the empirical data [224]. Implementing such a data-driven refinement loop would provide theoretical possibilities that were previously unfeasible through manual analysis alone.



## 6.8. Conclusion

This systematic mapping study has provided a comprehensive overview of research on dropout in tertiary-level continuing professional development (CPD) e-learning courses, synthesising 216 articles published between 2007 and 2023. The review offers three main contributions to the field.

First, by applying the mapping methodology, we have charted the research landscape, identifying both well-researched areas and notable gaps. The analysis revealed that research is heavily concentrated on MOOCs and is mainly based on quantitative methods, with limited attention to qualitative approaches, theoretical development, and intervention testing.

Second, the review has clarified the complex nature of dropout measurement in this context. Reported dropout rates vary considerably across studies, and the choice of baseline, such as enrolments versus active starters, affects interpretation. Several articles argue that calculations based on active starters provide a more meaningful picture, as they exclude individuals who never engaged with course content. However, our findings indicate that MOOC dropout remains high regardless of the calculation method used. Furthermore, the traditional notion of dropout may not fully capture the learning behaviours of CPD learners, many of whom engage selectively to meet specific professional goals rather than aiming for formal course completion.

Third, the synthesis of dropout and persistence factors across the literature has highlighted a consistent pattern: time constraints emerge as the primary barrier to completion, while motivation serves as the main driver of persistence. However, qualitative studies corroborated that while time limitations are the most cited reason, they might sometimes mask underlying issues such as motivation or content difficulty. Among the interventions studied, simple and well-timed approaches, such as personalised encouragement and clear guidance, often showed more consistent positive effects than more complex strategies like gamification or adaptive learning paths.

The review has several limitations that should be acknowledged. As is typical for mapping studies, we did not assess the methodological quality of individual articles and the synthesis reflects patterns in the literature rather than weighted findings. The search strategy, which focused on the term “dropout” across the full article texts, may have missed studies using alternative terminology, although the substantial number of retrieved articles suggests reasonable coverage. Additionally, the findings are shaped by the characteristics of the included literature, which is dominated by MOOC studies from European and North American contexts, and may not fully generalise to other forms of CPD or to learners in other regions.

Several directions for future research emerge from this review. Only two of the 216 articles called for new theoretical work, which we see as a clear weakness of the literature. The most pressing need is therefore for more theoretical development; this would help advance research beyond its current descriptive and predictive focus. The field would also benefit from greater methodological diversity, including more qualitative and mixed-method studies to complement the dominant quantitative tradition. There is a need for studies that extend beyond MOOCs to examine dropout in university-based CPD courses, commercial e-learning, and shorter course formats. The work on interventions would benefit from more structured experimental designs, and future studies should adopt more consistent reporting standards to allow better comparison and synthesis across the field.

Recent developments in generative AI and explainable AI (XAI) offer new ways to address the theoretical gap noted above. These tools can support automated theory synthesis across large bodies of literature, help uncover causal mechanisms that are difficult to detect manually, and enable iterative model building in which hypotheses are refined against empirical data. With appropriate human oversight, such approaches could help advance theoretical work on dropout in CPD e-learning.

## CRediT authorship contribution statement

**Panu Korpela:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Lauri Malmi:** Writing – review & editing, Validation, Supervision, Methodology, Formal analysis, Conceptualization.

## Declaration of Generative AI and AI-assisted technologies in the writing process

During the final stages of this work, the author(s) used Writefull and ChatGPT (OpenAI) to improve readability, wording, and language clarity, as well as to suggest  $\text{\LaTeX}$  code using the TikZ package for figures. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix. Supplementary material

### A.1. Bibliographic analysis

See [Table 20](#).

### A.2. Theories

**Categorisation of dropout (or persistence) reasons** In two articles, the models were used to identify and categorise factors influencing dropout or persistence, but the models do not emphasise interactions between factors. In [9] there is clear categorisation of dropout/persistence factors into individual characteristics, external factors, and internal motivational factors (such as satisfaction and relevance). Lastly, in [103] patterns of engagement and dropout in MOOCs are identified, a theoretical model of interacting factors is not presented.

**Structural equation model of some aspects of dropout/persistence** Of the 216 reviewed studies, nine used structural equation modelling (SEM) to examine the relationships that affect persistence (or dropout) in e-learning [76,77,98–100,151,175,188,192]. Despite their small number, certain relationships were encountered more than once in those that were found to have predictive power. Those included *motivation* as the factor most frequently mentioned (in four models), *support and feedback* in three, *relevance* in two, and *quality of environment* also in two.

Some SEM studies stood out for their complex frameworks. For example, in [99] a comprehensive model was used that incorporated motivational and support factors, moderated by demographics. They found that satisfaction, effort expectancy, motivation, performance expectancy, and support and feedback significantly predicted persistence. In [100], the focus of the

**Table 20**  
Articles with highest citation counts.

| Article name                                                                                                                               | First author            | Year | Citation count |
|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|------|----------------|
| Factors influencing adult learners' decision to drop out or persist in online learning [9]                                                 | Park, Ji-Hye            | 2009 | 1741           |
| Exploring the factors affecting MOOC retention: A survey study [71]                                                                        | Hone, Kate              | 2016 | 950            |
| Dropout rates of massive open online courses: behavioural patterns [12]                                                                    | Onah, Daniel            | 2014 | 761            |
| "Turn on, tune in, drop out": Anticipating student dropouts in massive open online courses [56]                                            | Yang, Diyi              | 2013 | 624            |
| Learning Engagement and Persistence in Massive Open Online Courses (MOOCS) [144]                                                           | Jung, Yeonji            | 2018 | 607            |
| Understanding Student Motivation, Behaviors and Perceptions in MOOCs [157]                                                                 | Zheng, Saijing          | 2015 | 584            |
| Dropout prediction in MOOCs using learner activity features [229]                                                                          | Lykourantzou, Ioanna    | 2009 | 557            |
| Sentiment Analysis in MOOC Discussion Forums: What does it tell us? [55]                                                                   | Wen, Miaomiao           | 2014 | 508            |
| What makes a great MOOC? An interdisciplinary analysis of student retention in online courses [124]                                        | Adamopoulos, Panagiotis | 2013 | 472            |
| Attrition and achievement gaps in online learning [142]                                                                                    | Kizilcec, René          | 2015 | 465            |
| Discriminating factors between completers of and dropouts from online learning courses [165]                                               | Lee, Youngju            | 2013 | 437            |
| Predicting MOOC dropout over weeks using machine learning methods [230]                                                                    | Kloft, Marius           | 2014 | 414            |
| What predicts student satisfaction with MOOCs: A gradient boosting trees supervised machine learning and sentiment analysis approach [141] | Hew, Khe                | 2020 | 401            |

model was on psychological needs and engagement factors, of which support and feedback was found to be influential. Furthermore, in [188], insights from social exchange theory and gamification research were combined. In it, social interactions and gamification rewards enhanced learner motivation and course satisfaction, as well as having a good course description.

As a special mention, in [76] there is in itself no ranking of characteristics, which would be the most predictive of dropout. Instead, its primary objective was to develop and validate reliable instruments (scales) that could later be used to measure such predictors. One of the scales was for environmental factors and the other for persistence factors.

**Model with interaction between factors** The three articles in this category model interactions between factors, although they do not use a structural equation model (SEM) in the analyses. In [96], the effect of gamification strategies on engagement and completion rates in MOOCs is inspected through a model called E-MIGA (Integrated Theoretical Gamification Model in e-Learning Environments). In [117], self-regulated learning (SRL) was used to guide research where both self-reported and event-based analyses of SRL strategies are evaluated and compared to measure their effect on prediction of dropout. Lastly, in [124] an explanatory model was first created, which was then used as a basis for predictive modelling.

### A.3. Analysis methods

The **analysis methods** are depicted in Fig. 17. The description of the categories is below.

We classify the *quantitative* analysis methods as follows:

**qn\_descriptive\_statistics** The article uses, for example, percentages to describe how dropout happened from week to week in a course [42,149], or which percentages the questionnaire respondents' answers have [92,94,173]. In addition, reviews sometimes deal with the findings using descriptive statistics [21].

**qn\_advanced\_statistics** In this case, an article may have utilised regression to predict dropout from some variables [67,73,178]. Also cases where questionnaire results' significance was presented using, for example, chi-square analysis, it has been classified in this category [147].

**qn\_machine\_learning** Machine learning techniques related to dropout are used to predict the phenomenon from some available data [231–233]. We have not divided machine learning articles into further categories, as several such review articles already exist [15–17,125].

*Qualitative* analysis method categories are divided between:

**ql\_actual\_method** This category includes cases where the article had used some declared method for analysing the qualitative findings. Examples include, as reported in the articles themselves, ethnographic approach [104], thematic analysis [172], grounded theory method [157], or phenomenological method [79].

**ql\_samples\_only** Almost all articles in this category present samples from interviews without a more thorough analysis [71,94,171]. In one case the article includes discussion about course platform design [74] – it is counted here as the portion is declared in the paper to be a part of a mixed methods study.

### A.4. Data sources

#### A.4.1. Data sources/research setup

Regarding research setup, the articles analysed drew on a range of data sources, either from courses, pre-existing datasets, or participant surveys. Thus, the data sources for the articles were classified as follows.

**course** Researchers can directly observe the course being investigated. They may possibly influence what kind of data they get and may even change the technical setup of the e-learning system to get what they want. In addition, cases where the research team has been involved in creating the course have been marked as "course", even if they have not directly observed the execution [135,152].

**interview\_or\_questionnaire** The data given by the course or course management system is mainly a source for getting the dropout status, and the actual results have been gathered, for example, via interviews or questionnaires. Articles that considered the survey portion to be the main part of the study were also classified in this category (for example, [93, p. 7]), as well as articles that only had a survey portion without specific courses (for example, [157,172]).



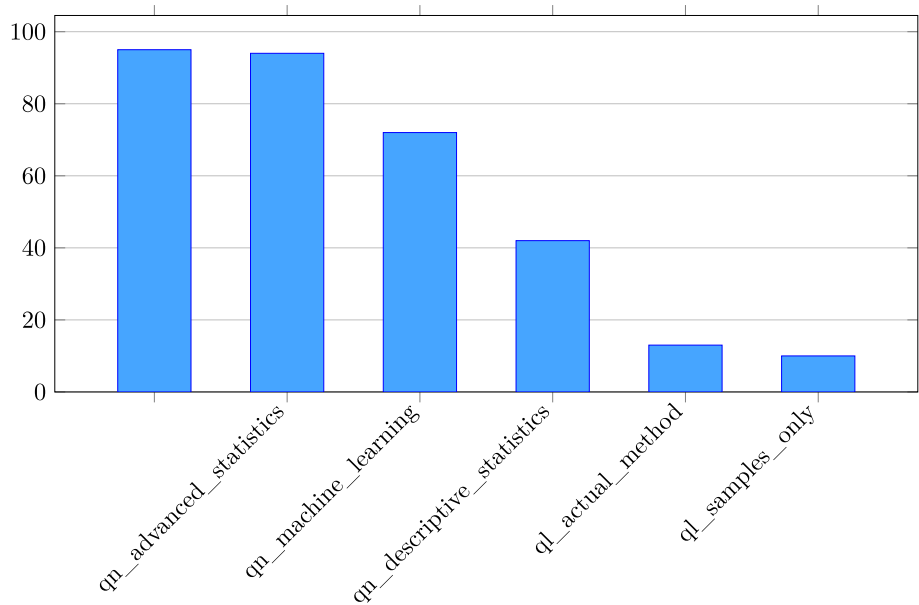


Fig. 17. Analysis methods.

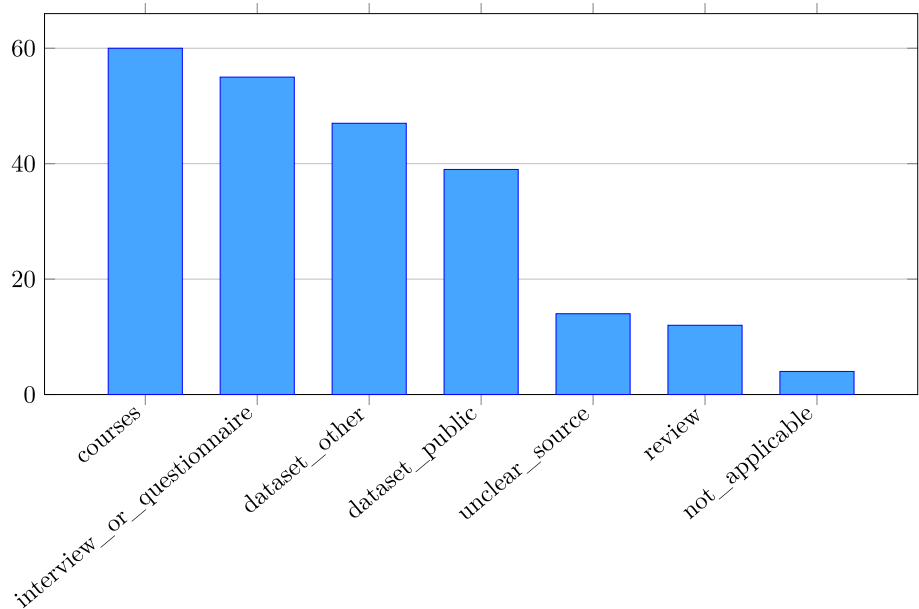


Fig. 18. Data sources.

**dataset\_public** This category covers pre-existing public datasets. Recent overviews of available datasets can be found in [234, 235]. Examples of common bigger dataset sources are KDD Cup 2015 [236], MITx/HarvardX [237], and Stanford [238]. Also some individual ones exist, such as “Computer science” course from Hellenic Online University [239].

**dataset\_other** Pre-existing, not public dataset. May, for example, include the database data of courses, which have afterwards been processed by the article authors (as examples, [81,184]). As a special mention, FutureLearn [240] is a MOOC platform that provides partner institutions with access to data related to the courses they run [103, p. 59], and as such FutureLearn courses are included in this category.

**review** A review article (examples: [14,23,25]), which can also be a mapping study [21].

**unclear\_source** This category includes cases where the data do not belong to the above categories. In some cases, the course(s) being analysed are described, but there is no mention of how the data have been obtained [42,80,197]. Or there may be little or no mention of the source at all [84,241].

**not\_applicable** The article does not analyse course data, as it contains only dropout-related remedies [112,242], a replication of an earlier study [110], or uses simulated data [193].

The counts of the data source categories are shown in Fig. 18.

**A.4.2. Data sources/Kinds of data used**  
Various **kinds of data** were used to analyse dropout or persistence. These have been classified in Table 21; the counts are shown in the same table. If some information had been collected in an article – most

**Table 21**  
Kinds of data utilised.

| Category                 | Description                                                                                                                                                                                                     | Count |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| activities_recorded      | Course activity data on a higher level. Includes, for example, quiz or forum participation. Examples: [54,101,181]. In most cases, this category is also combined with other kinds of data (e.g. [58,109,115]). | 126   |
| Clickstream              | Low-level actions such as keystroke and clickstream data. Amount and timing of clicking a quiz are also in this category. Examples: [134,230,243]                                                               | 112   |
| Questionnaire            | Questionnaire data, which may include fixed or open-ended questions. Examples: [98,188,204].                                                                                                                    | 70    |
| course_results           | Grades from examinations, homework or assignments. Example: [170]. In all other cases this category is combined with other categories. [154,164,244].                                                           | 60    |
| Demographics             | Participants' demographic data, if it has been analysed in relation to dropout or persistence [9,75,156].                                                                                                       | 44    |
| descriptive_demographics | Participants' demographic data, if it only describes the data set (e.g. [105,128]).                                                                                                                             | 28    |
| not_applicable           | Reviews, theory-building articles. Also simulations are included in this category [193].                                                                                                                        | 13    |
| Interview                | In-person interviews. Examples: [79,104,157].                                                                                                                                                                   | 10    |
| Other                    | Feedback given by the students about a course [141], and course data created by a computer simulation [242]                                                                                                     | 7     |

**Table 22**  
Course subject classification.

| Category                           | Example course titles                                                                                                                                                  | Count |
|------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 00 generic programmes              | Developing your research project [81], Etiquette [245]                                                                                                                 | 4     |
| 01 education                       | Cloud based learning 2013 [50], Applied educational innovation (AEI) [86], Problem based learning [195]                                                                | 31    |
| 02 arts humanities languages       | Introduction to art: concepts and techniques [187], The modern world: global history since 1760 [163], Greek heroes [97]                                               | 30    |
| 03 social sciences                 | ePSYx trends in e-psychology [197], Discovering political science [154], Principles of microeconomics [152]                                                            | 18    |
| 04 business administration law     | Project management [241], Insurance intermediaries' certification course [102], Foundations of business strategy [163]                                                 | 35    |
| 05 STEM                            | The fascination of crystals and symmetry [167], I heart stats [194], College algebra and problem solving [208]                                                         | 34    |
| 06 ICT                             | Introductory programming in Java [191], DevOps: Competences for smart cities [246], Introduction to programming [247]                                                  | 71    |
| 07 engineering                     | Nanotechnology and nanosensors [10], Electrons in action [117], Mechanical behavior of materials: beams, stress transformations, columns, and cellular materials [170] | 21    |
| 08 agriculture                     | "agriculture" given as a field [84]                                                                                                                                    | 3     |
| 09 health                          | Dr Internet [149], Medical urgencies [48], Literature and mental health [60]                                                                                           | 14    |
| 10 services                        | Introduction to halal food [127]                                                                                                                                       | 1     |
| Not applicable                     | Includes reviews, simulations, and theory building articles                                                                                                            | 24    |
| Unclear/not mentioned <sup>a</sup> | Android 2013 [50], Inserlab [126], MOOC CH50x [63]/[9,172,248]                                                                                                         | 50    |

<sup>a</sup> Meaning that the contents of the course cannot be deducted from the information provided. For example, just the name "Android 2013" does not give any information about the scope of the course. This category covers also cases where the subject has not been mentioned at all.

notably demographic data – but not used for the dropout analysis, it would not have been included in the respective category.

Of the classifications that were revealed, "clickstream" and "activities\_recorded" are typically rather easy to extract from an e-learning system. The first one is typically relatively easy to process by a computer, the second one may need more preprocessing. Using "clickstream" data is understandably limited by what has been stored from using the e-learning system: is it only page transitions and video views, or something more granular [249].

In line with a large majority of articles having quantitative research designs, "activities\_recorded", "clickstream", "course\_results", and both versions of "demographics" constitute 79.0% of *all classified data sources*. As a separate category, "questionnaire" has been used relatively often with a share of 32.4% – they can of course consist of quantitative or qualitative portions or a combination of both.

Demographic data varies greatly; typically, it may contain age information, geographic location, gender, educational level, or occupational status. In some cases demographic data were analysed in relation to dropout, in other it was just used to describe the setting.

Using in-person interviews was surprisingly rare, only 10 or 4.6% of the articles had decided to use them as a data source.

#### A.5. Course subjects, educational levels, workloads, and participant profiles

The **course subjects** were classified according to the *ISCED categorisation of the fields of education and training* [46]. Classification was performed only on courses that did not depend on public datasets. Articles using such datasets were omitted as in very many cases they focused on dropout prediction via machine learning, and the dataset was only a source of data for the mathematical treatment, and as such the course subjects of the dataset carried no meaning.

It is worth noting that course subject categorisation is descriptive at best: some articles only have single participations on any given course [104], some even had tens of thousands [58,90].

In many cases, only course categories were given [129,142,250] – mapping them to the ISCED educational field categorisation was relatively straightforward.

The subject classes and their brief descriptions are listed in Table 22.

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